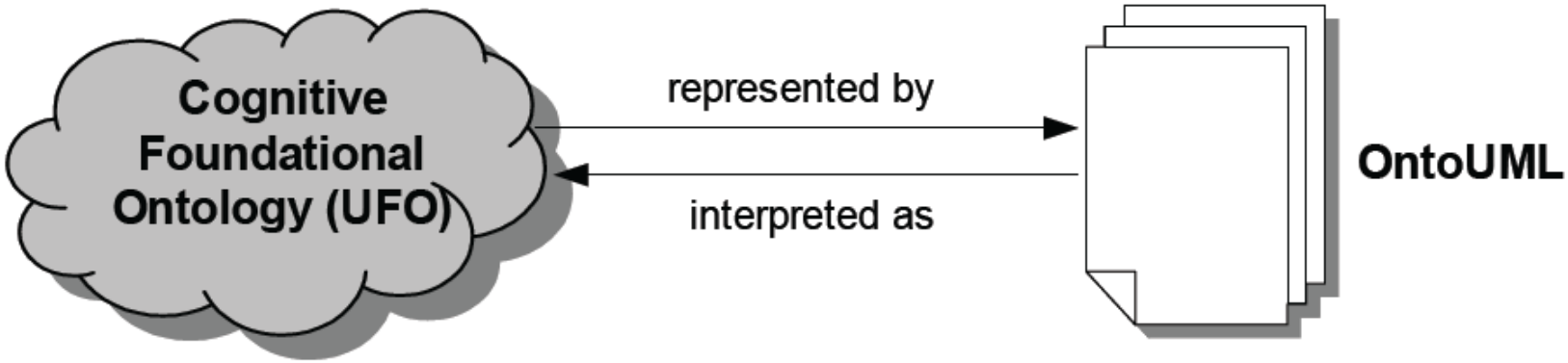
Unified
Foundational
Ontology

ONTOUML



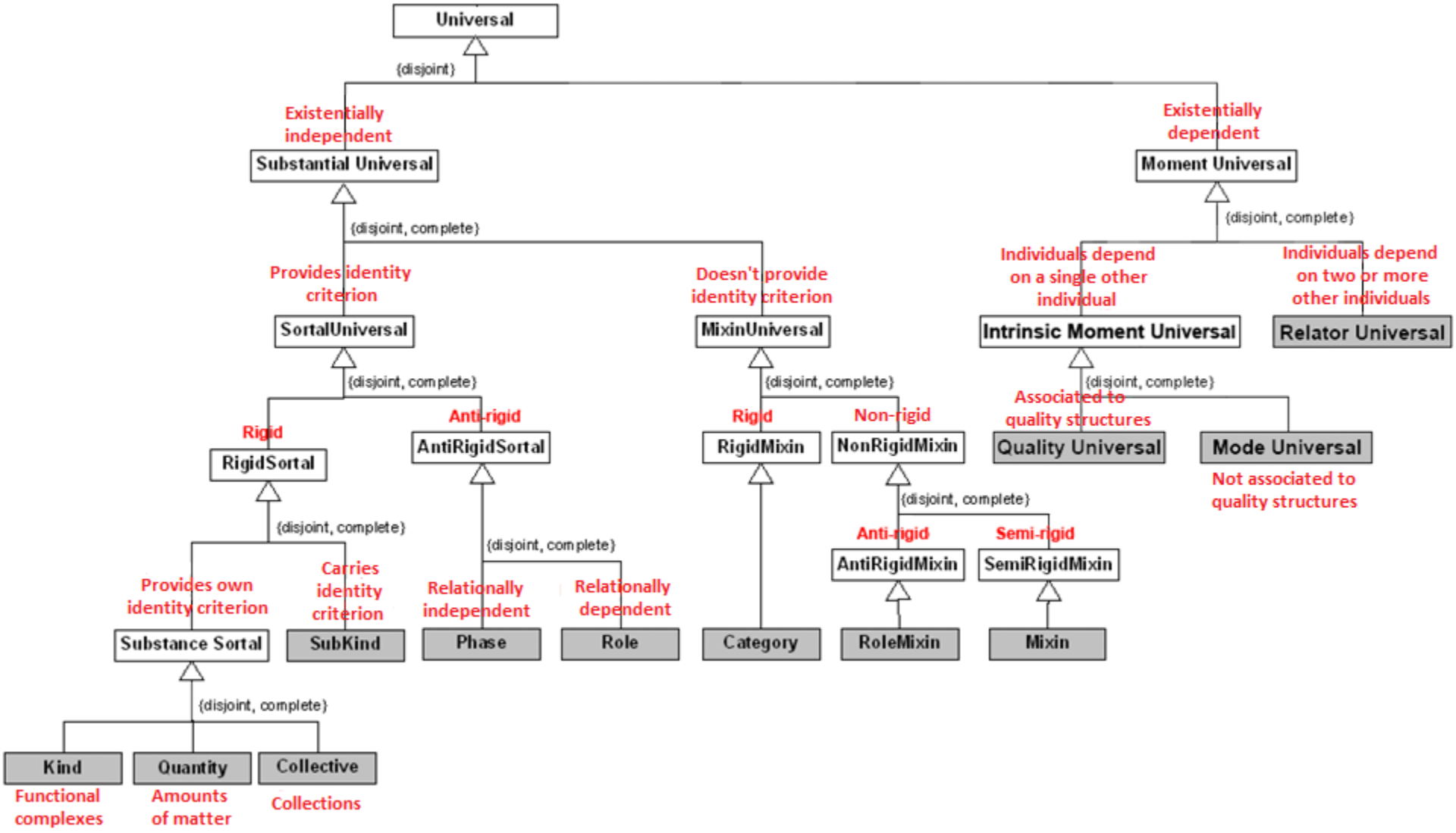
The slides about UFO were modified from Giancarlo Gizzardi

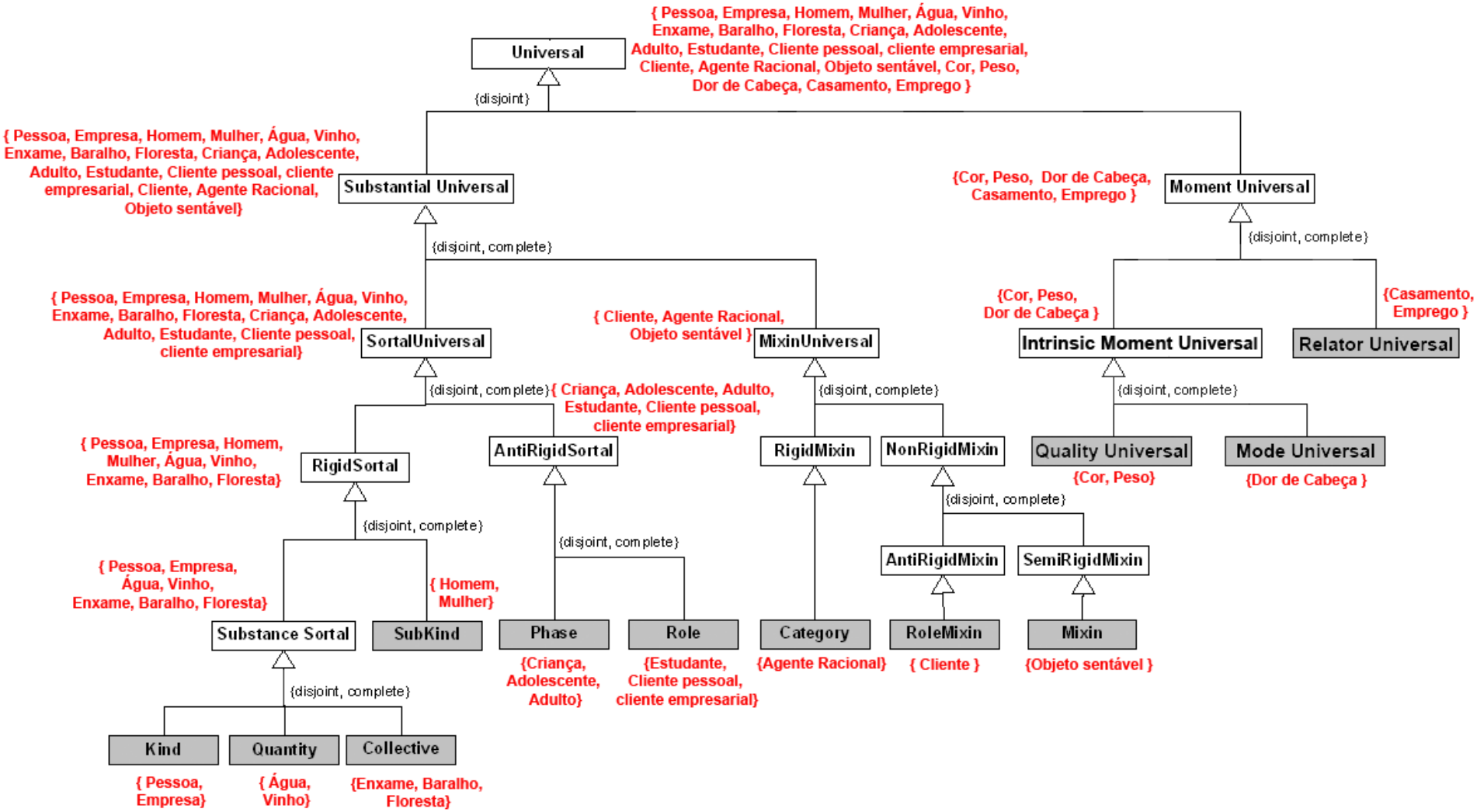
Giancarlo Gizzardi

Universal, Class, Particular, Instance

- ***Universal*** is characteristics that particular things have in common, In other words, universals are **repeatable** or recurrent entities that can be instantiated or exemplified by many particular things. It is an ***intensional*** definition; it is a description that is sufficiently generalized to fit every member of a set.
  - Suppose there are two chairs in a room, each of which is green. These two chairs both share the quality of "chairness", as well as greenness or the quality of being green; in other words, they share a "universal".
- ***Class*** is a set in which an individual member can belong for one of the two ways:
  - it is included in an extensional list of members (enumeration);
  - it matches an intentional definition of one set member.
- ***Particular*** is the concrete entity that holds the properties of the Universal, i.e, it is the instance.

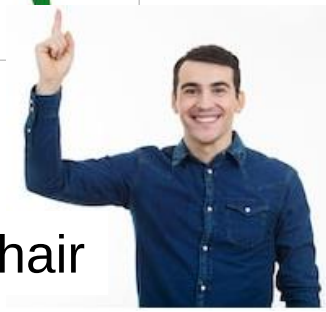
UFO A Meta-types for Continuants




Universal
{ Pessoa, Empresa, Homem, Mulher, Água, Vinho, Enxame, Baralho, Floresta, Criança, Adolescente, Adulto, Estudante, Cliente pessoal, cliente empresarial, Cliente, Agente Racional, Objeto sentável, Cor, Peso, Dor de Cabeça, Casamento, Emprego }
{disjoint}
Substantial Universal
{ Pessoa, Empresa, Homem, Mulher, Água, Vinho, Enxame, Baralho, Floresta, Criança, Adolescente, Adulto, Estudante, Cliente pessoal, cliente empresarial, Cliente, Agente Racional, Objeto sentável}
Moment Universal
{Cor, Peso, Dor de Cabeça, Casamento, Emprego }
{disjoint, complete}
SortalUniversal
{ Pessoa, Empresa, Homem, Mulher, Água, Vinho, Enxame, Baralho, Floresta, Criança, Adolescente, Adulto, Estudante, Cliente pessoal, cliente empresarial}
MixinUniversal
{ Cliente, Agente Racional, Objeto sentável }
Intrinsic Moment Universal
{Cor, Peso, Dor de Cabeça }
Relator Universal
{Casamento, Emprego }
{disjoint, complete}
RigidSortal
{ Pessoa, Empresa, Homem, Mulher, Água, Vinho, Enxame, Baralho, Floresta}
AntiRigidSortal
{ Criança, Adolescente, Adulto, Estudante, Cliente pessoal, cliente empresarial}
RigidMixin
NonRigidMixin
Quality Universal
{Cor, Peso}
Mode Universal
{Dor de Cabeça }
Substance Sortal
{ Pessoa, Empresa, Água, Vinho, Enxame, Baralho, Floresta}
SubKind
{ Homem, Mulher}
Phase
{Criança, Adolescente, Adulto}
Role
{Estudante, Cliente pessoal, cliente empresarial}
Category
{Agente Racional}
AntiRigidMixin
SemiRigidMixin
RoleMixin
{ Cliente }
Mixin
{Objeto sentável }
Kind
{ Pessoa, Empresa}
Quantity
{ Água, Vinho}
Collective
{Enxame, Baralho, Floresta}

1. Meta-Types
Sortal and Mixin Universals

- Mixin universals meet the application principle for the individuals they collect
- Sortal universals meet the application principle and the identity principle



Green chair

Green seats

Green bench

Axiom

- (1) Declarations of identity and individuation can only be made for a SORTAL, which provides the identity and individuation to the individual it collects.

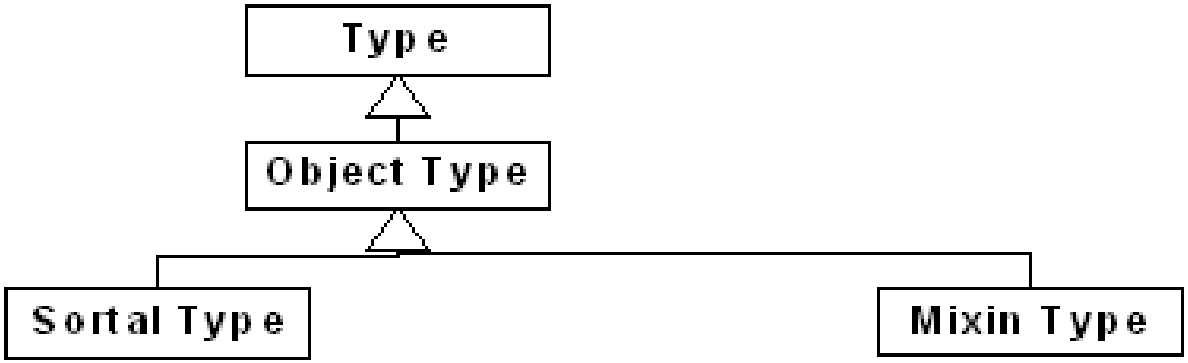
*Each object in a domain Conceptual Model must be an instance of a **type** of the model that represents a sortal*

Axiom

- (2) An individual cannot obey to incompatible identity principles.

Is ocean a kind of water?

1 Meta-Types



{Person, Apple, Student}

{Insurance item, red, tool }

Properties of a Sortal Universal

- Countability (unity):
  - An object is countable if whenever it represents an object X, it does not represent a connected part of X
  - Ex. *wood* is uncountable, since a part of a piece of wood is still a piece of wood (homeomerous)
- Temporal stability
  - An object is stable in time if whenever it represents an object X at time 1, it will represent X at time N
  - Ex. *Dog* is temporally stable, while *Puppy* is not

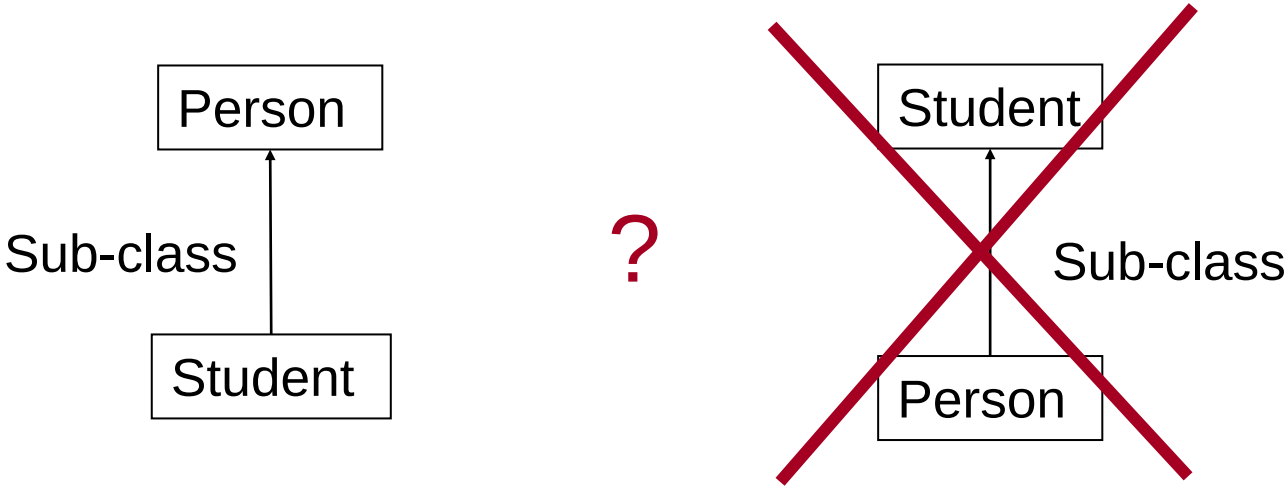
Unique Identity Principle

Temporal stability
(diachronic identity)

Puppy

Dog

Rigid Property (R+)



- A type T is rigid if for any X instance of T, X will necessarily be instance of T in all possible worlds.
- An instance of student can stop being a student without losing it identity. An instance of person cannot stop being a person.

Anti-Rigid Property (R~)

- A type T is anti-rigid if for any instance X of T, X may not be an instance of T (in modal sense).
- There is a world where X is not an instance of T.
  - *Pet, student, professor, a transport , a container...*

Object Properties

- Rigid properties (essential to all instances) +R
  - Being a person = true
  - Every instance of a person will be always a person
- Semi-rigid properties -R
  - Being hard
  - Is essential to some instances (hammer), but not to others (fruit)
- Anti-rigid properties ~R
  - Being a husband
  - Every instance of a husband were not or can stop be a husband

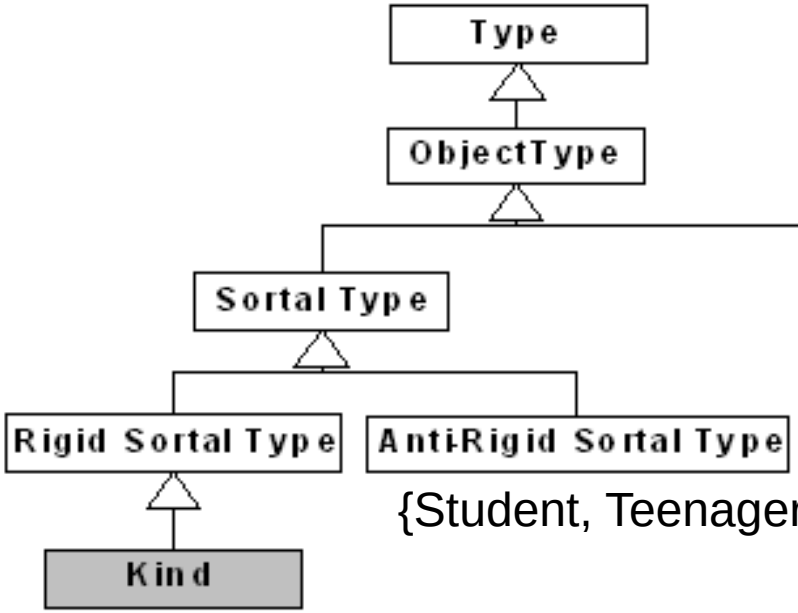
Rigid properties cannot be specializations of anti-rigid properties.

Person cannot be specialization of Student

Guarino, N and Welty, C. *Evaluating ontological decisions with Ontoclean.* Comm of ACM, V.45, n2, Feb, 2002.

1. Meta-Types

or Non-sortal Type



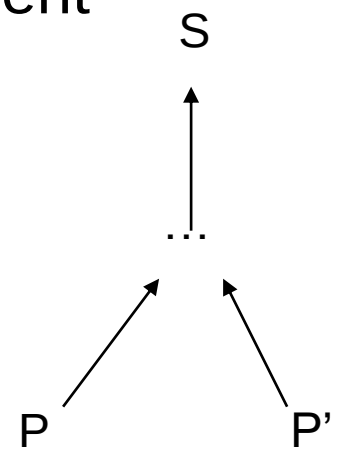
{insurable item, red}

{Student, Teenager, football player}

{Person, Apple, Enterprise}

Axiom

- (3) If an individual specializes two sortals throughout its existence, there must be exactly *one* rigid sortal that is super-class of these two sortals, of which both inherit the identity principle.



Ontological Identity X Identity

- **Provides Ontological Identity** (**+O**)

– Has an identity criterion or essential property that was not inherited

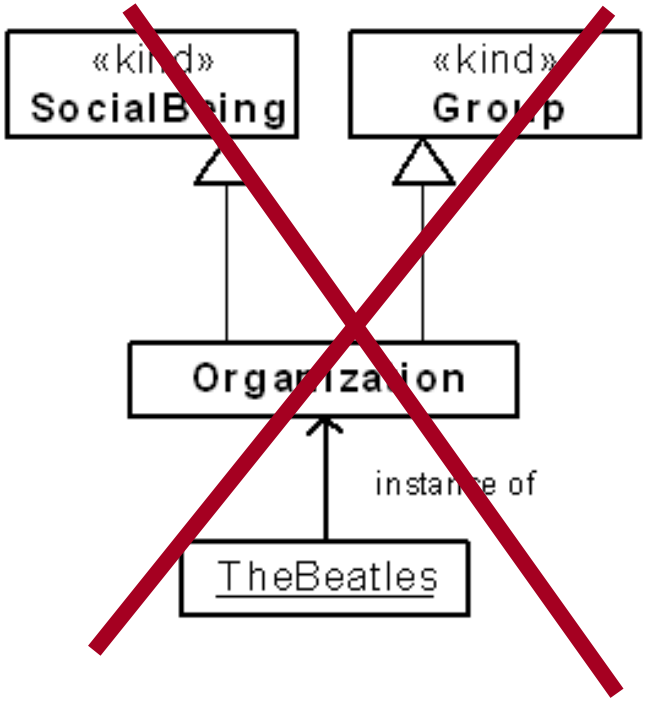
- **Carries Identity** (**+I**)

– Inherits its individualizing property.

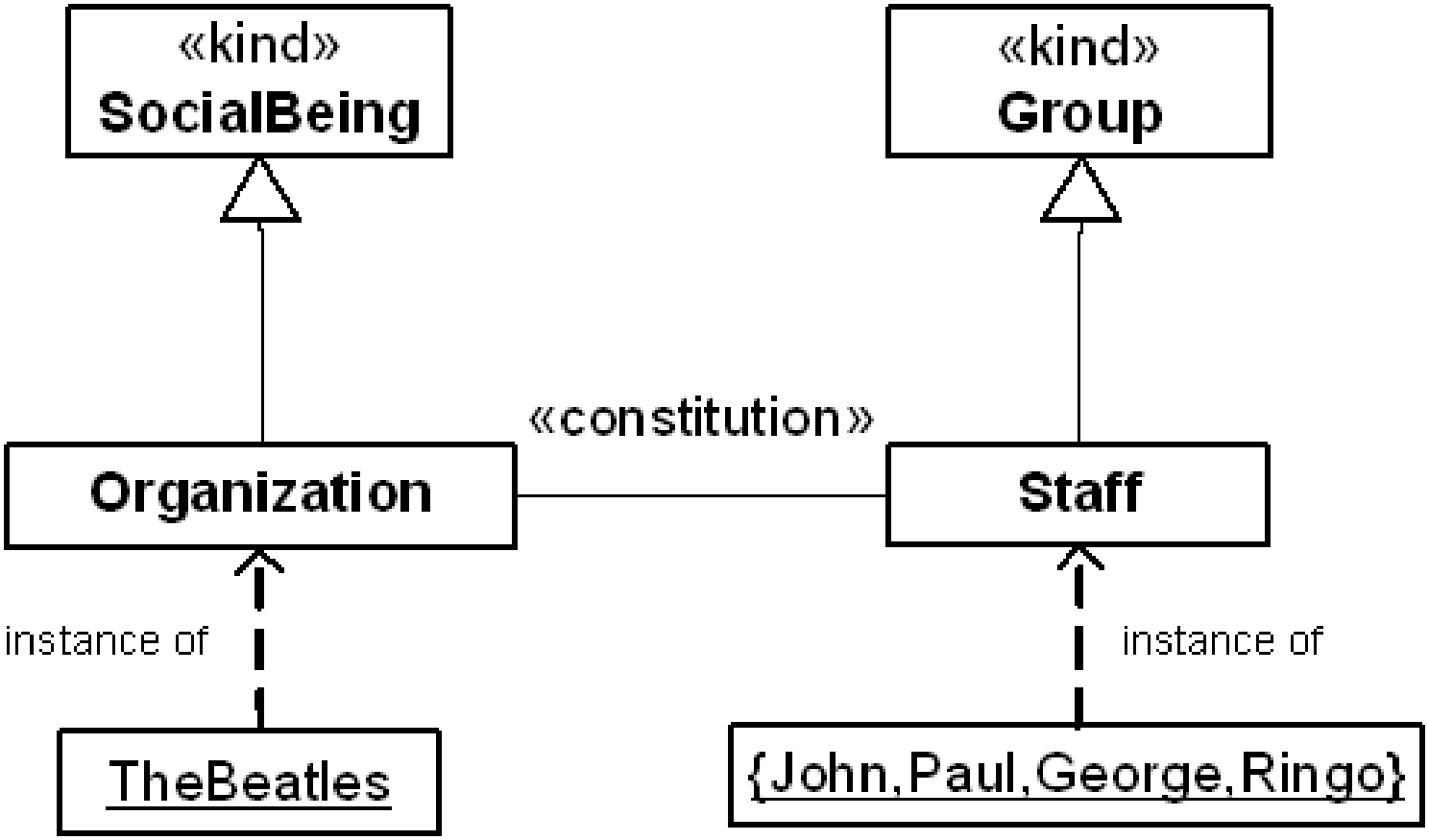
- Only rigid properties provide ontological identity
- Rigid properties (essential for all its instances) = +R
  - Person.born = true

Fundamentals

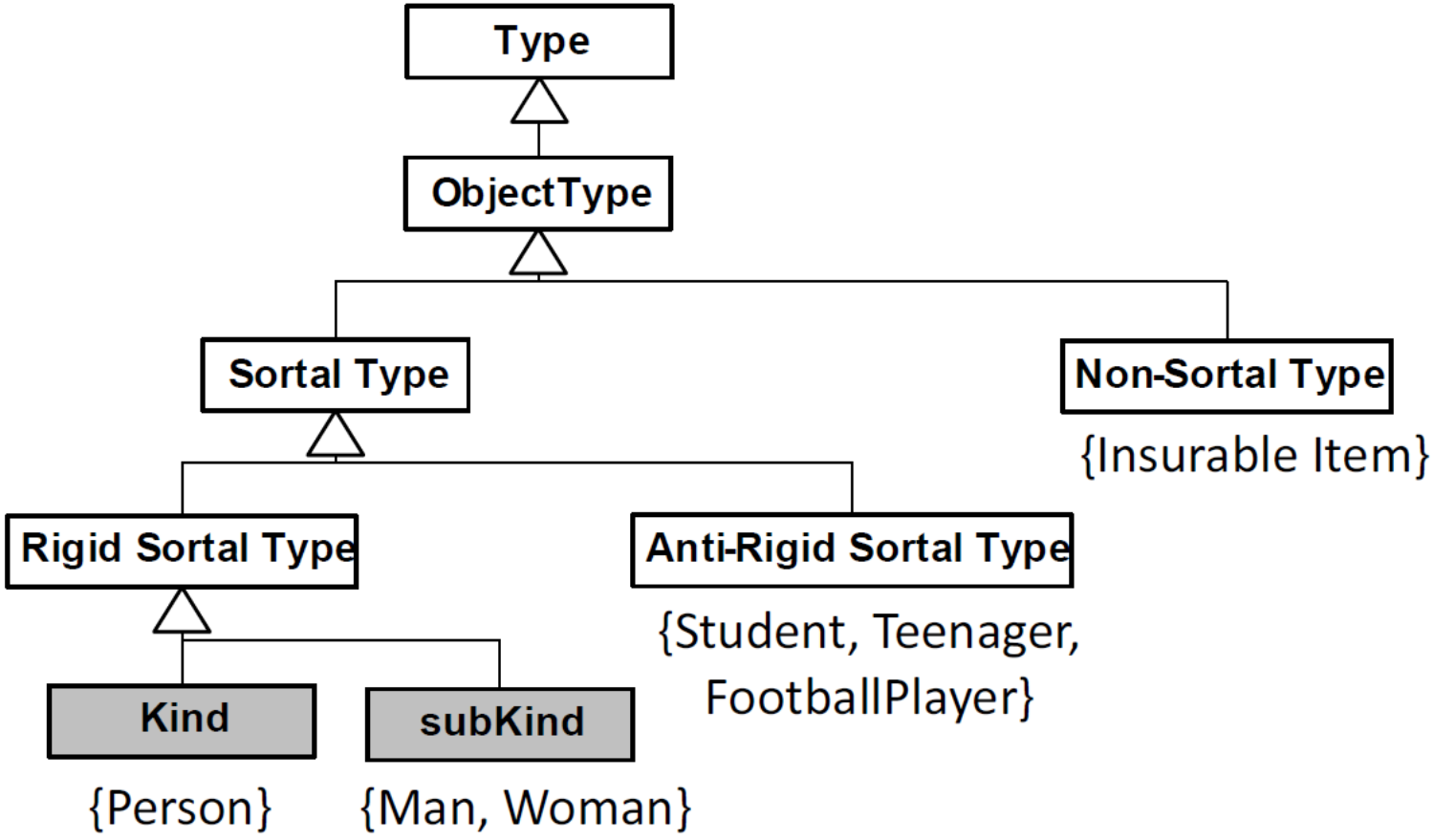
- A non-sortal type cannot have direct instances.
- A sortal type cannot appear in the model as a subtype of a non-sortal.
- An object in a model cannot be a direct instance of more than one TYPE (multiple inheritance is not admissible)
- A kind can not be a supertype of other kind.



- The Beatles are an organization that is both a social being and a group.
- Social being allows one of its members to leave and remain the *same* social being.
- If a member of a group leaves, the group *is not* the same.
- Incompatible identity principles!!


«kind»
SocialBeing
«kind»
Group
«constitution»
Organization
Staff
instance of
instance of
TheBeatles
{John,Paul,George,Ringo}

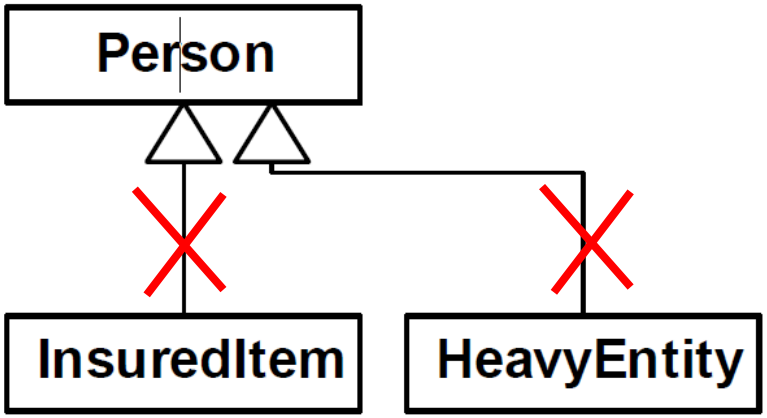
1. Meta-Types


Type
ObjectType
Sortal Type
Non-Sortal Type
{Insurable Item}
Rigid Sortal Type
Anti-Rigid Sortal Type
{Student, Teenager, FootballPlayer}
Kind
subKind
{Person}
{Man, Woman}

Foundations

Since the unique principle of identity supplied by a Kind is inherited by its subclasses, we have that:

- A Non-sortal type cannot appear in a conceptual model as a subtype of a sortal



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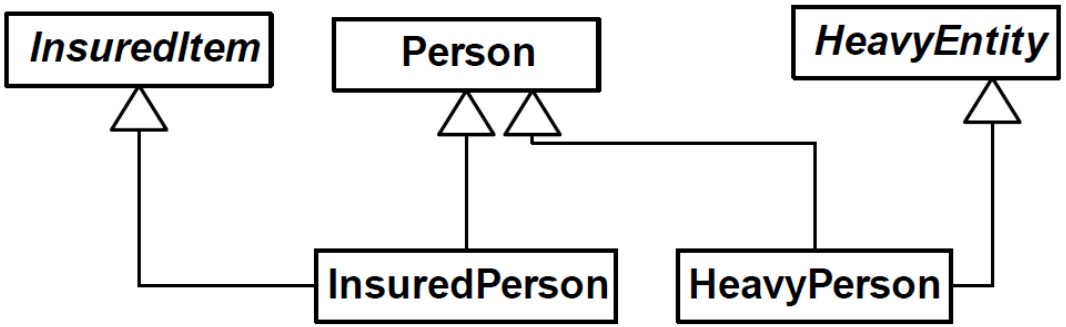
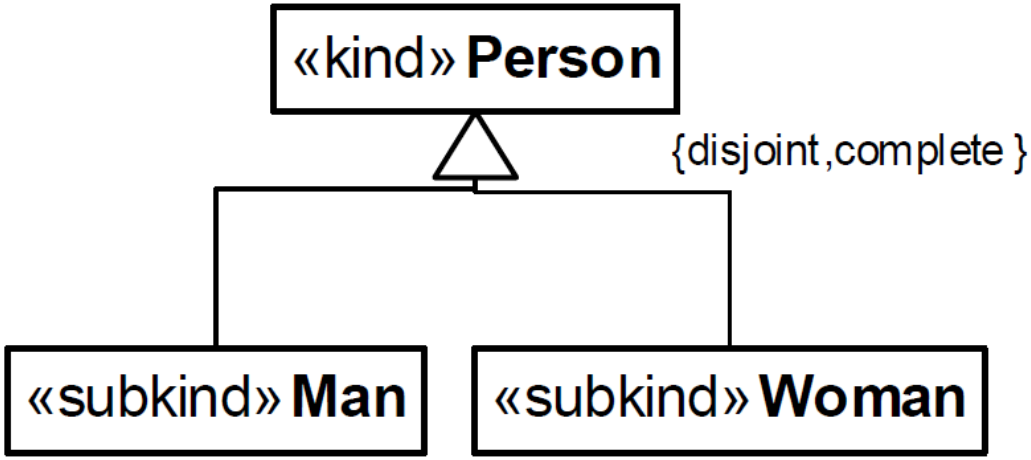


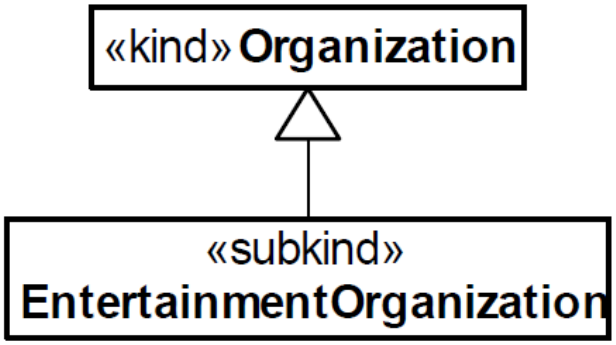
Subkind partitions

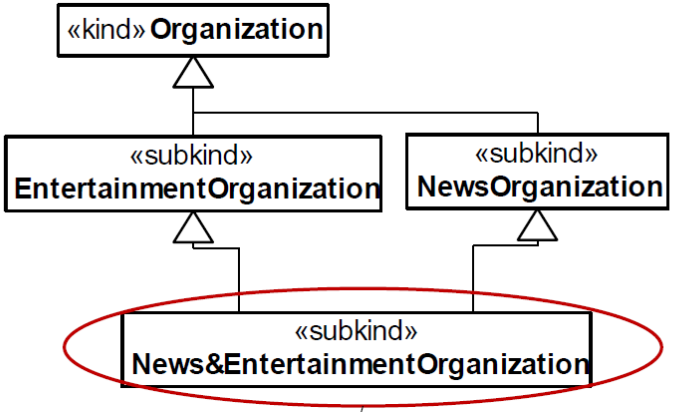
- Typically, subkinds define disjoint partitions over the particulars that are true in every possible world. But not necessarily complete.



Subkind

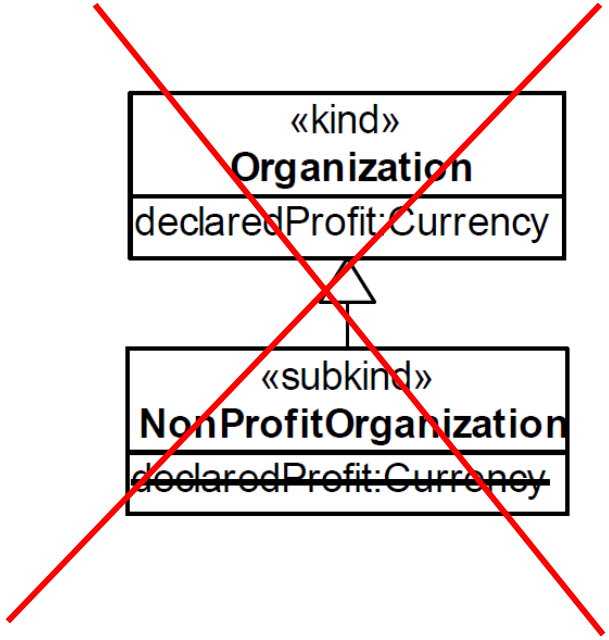
Subkinds can define specializations of kinds if the properties are genuine (intrinsic or relational) and rigid

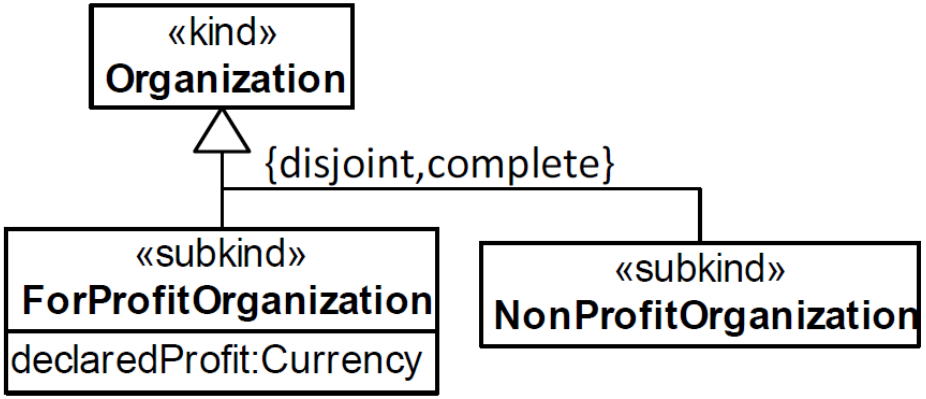




Subkind

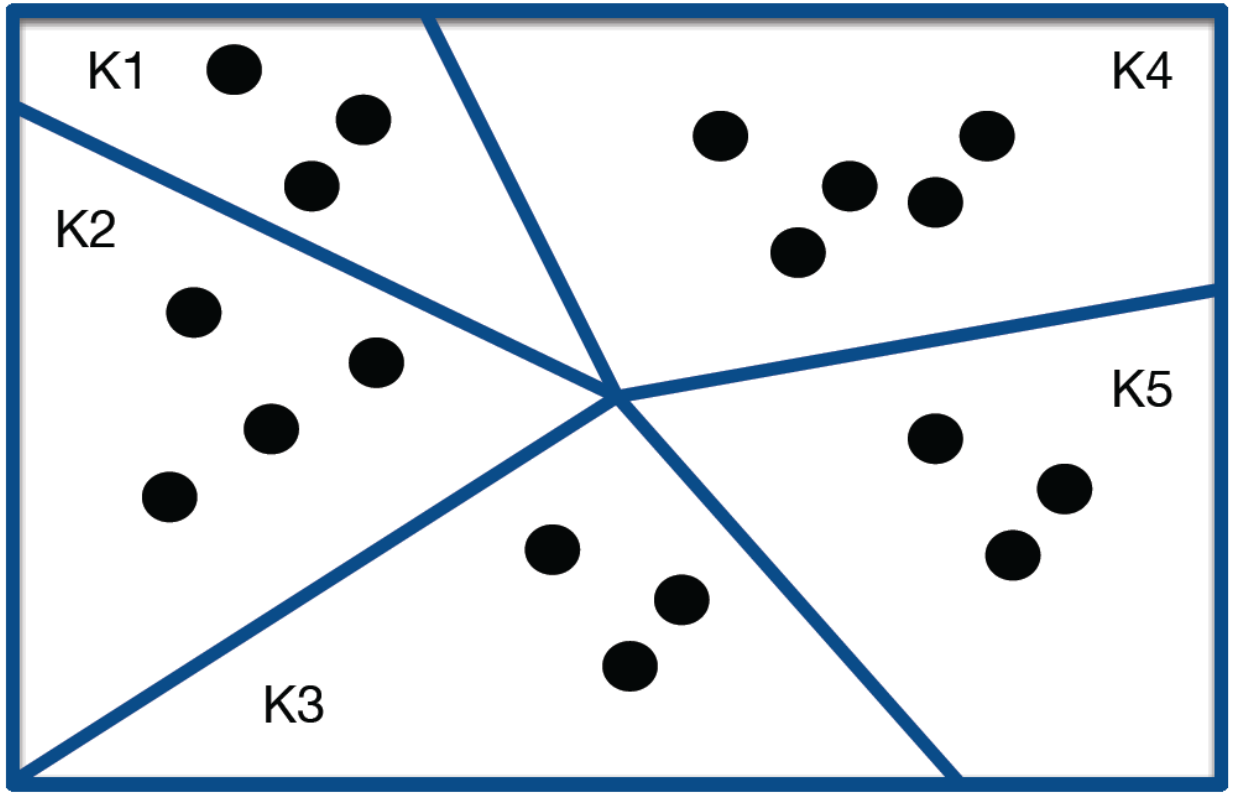
Remember that property overriding is always a bad idea and it is always caused by a conceptual modeling mistake

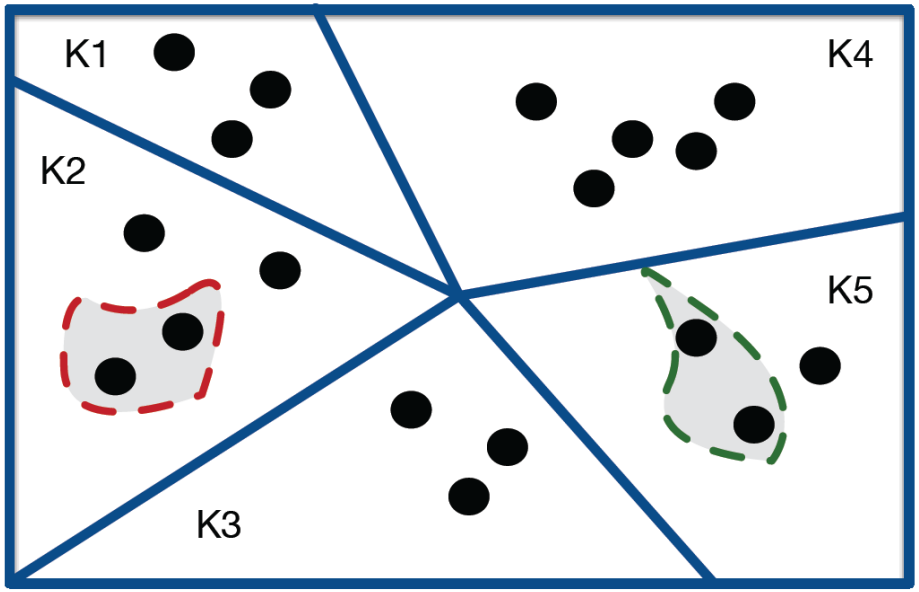




Kinds

Partition based on essential rigid properties

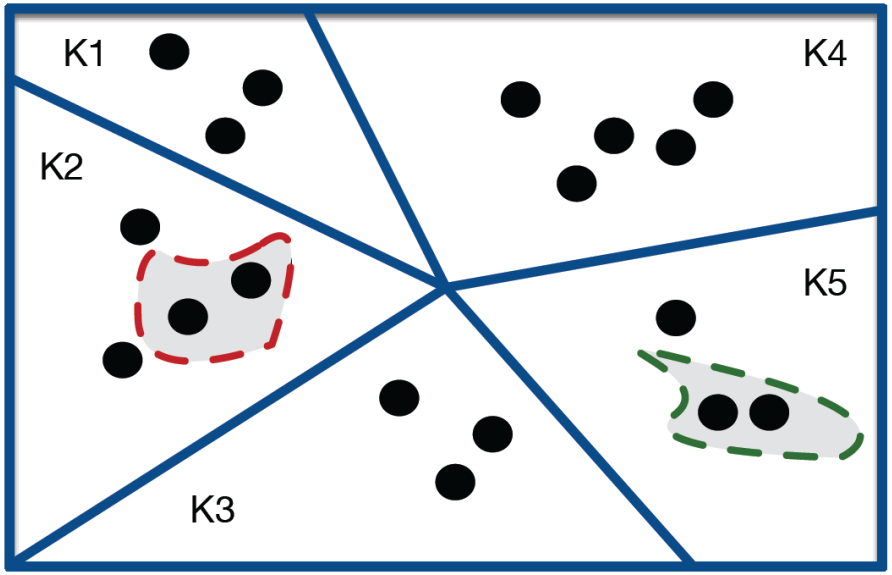




Anti-rigid sortals

Phase and role

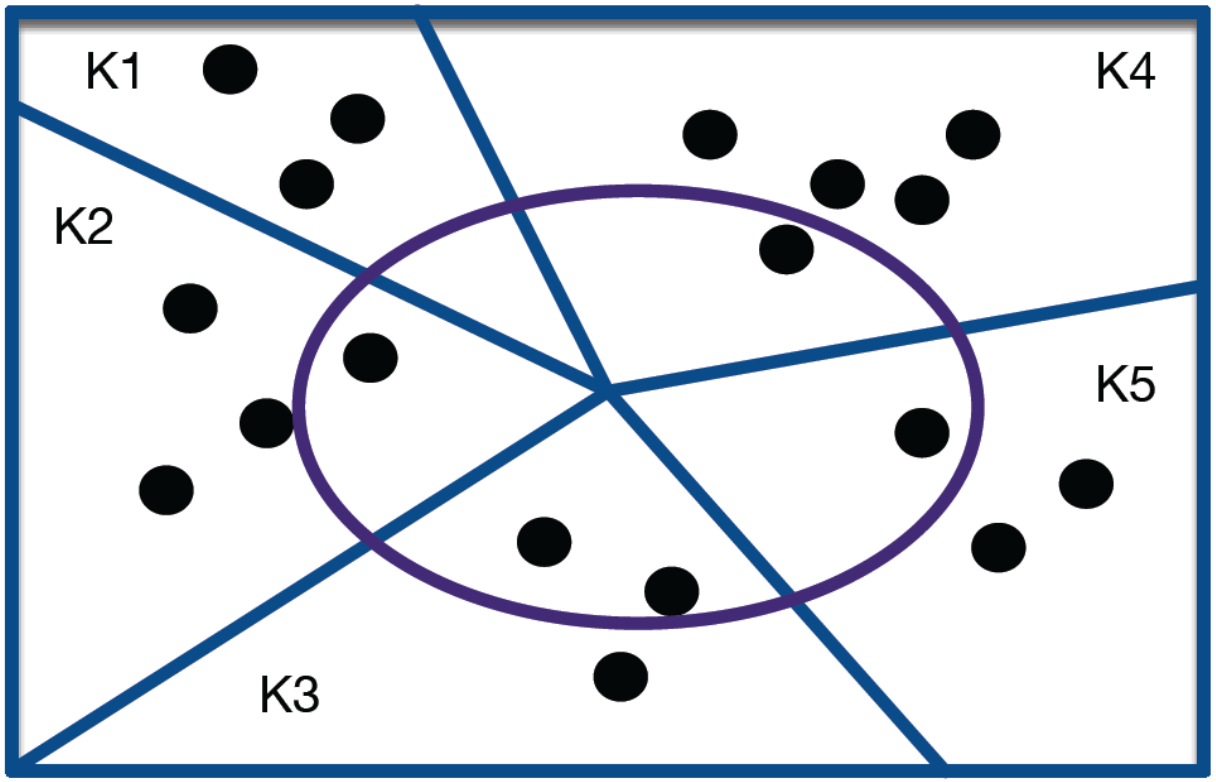
Partitions based on anti-rigid non-essential properties



Non-sortals

Partition based on common shared non-essential properties among distinct sortals

Mixin
Role-mixin
Categories

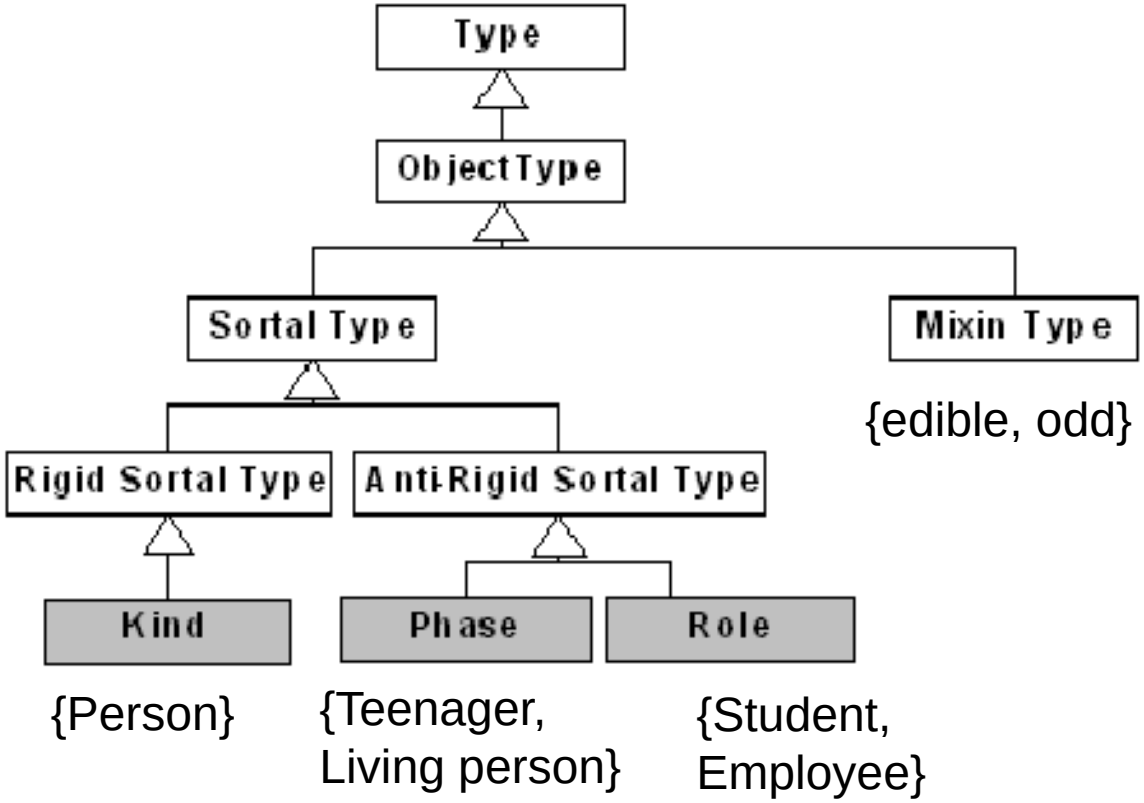


Quantity

- Maximally connected amount of matter
- Rigid, carries and supplies identity,
- Doesn’t meet the individuation principle (uncountable)
- *Homeomerous:* an individual whose parts are of the same type, that is, can be subjected to infinite decomposition without loss of identity
- Water, wood, sediment..

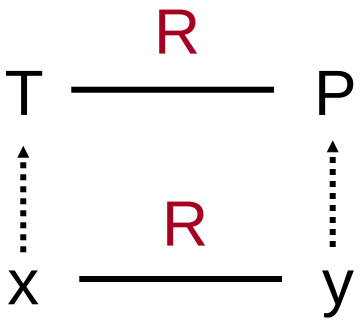
Guizzadi, Thesis, p 174

1. Meta-Types


Type
ObjectType
Sortal Type
Mixin Type
{edible, odd}
Rigid Sortal Type
AntiRigid Sortal Type
Kind
Phase
Role
{Person}
{Teenager,
Living person}
{Student,
Employee}

Relational Dependent Property (D+)

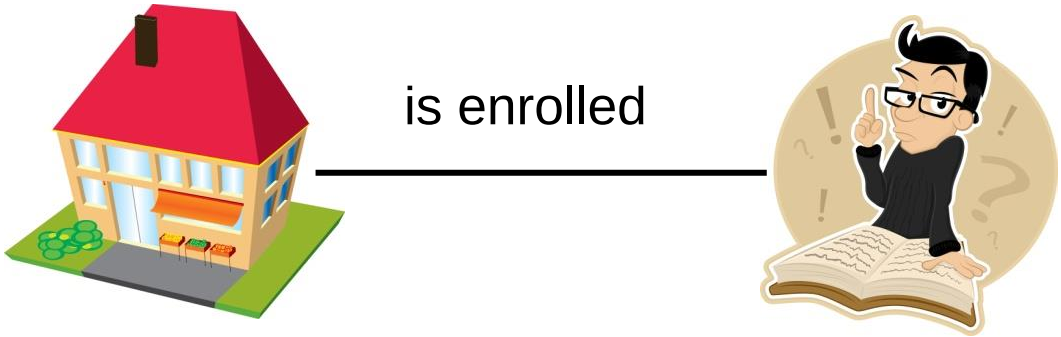
- A type **T** is relational dependent of another type **P** through a relation **R** if for any instance **x** of **T** there is an instance **y** of **P**, such that **x** and **y** are related by **R**.



There is no instance without the relation!

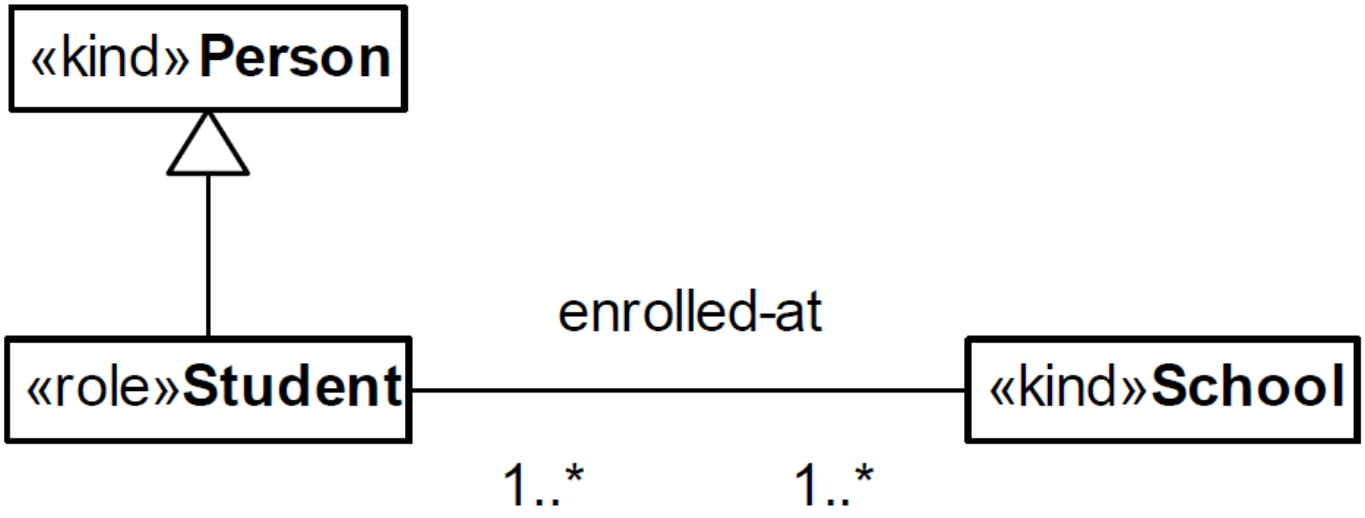
Roles

- Roles are relational dependent types defined as na anti-rigid specialization of a kind based in a relation
  - To define ***student***, there must be an ***educational institution*** with which it is related



Roles

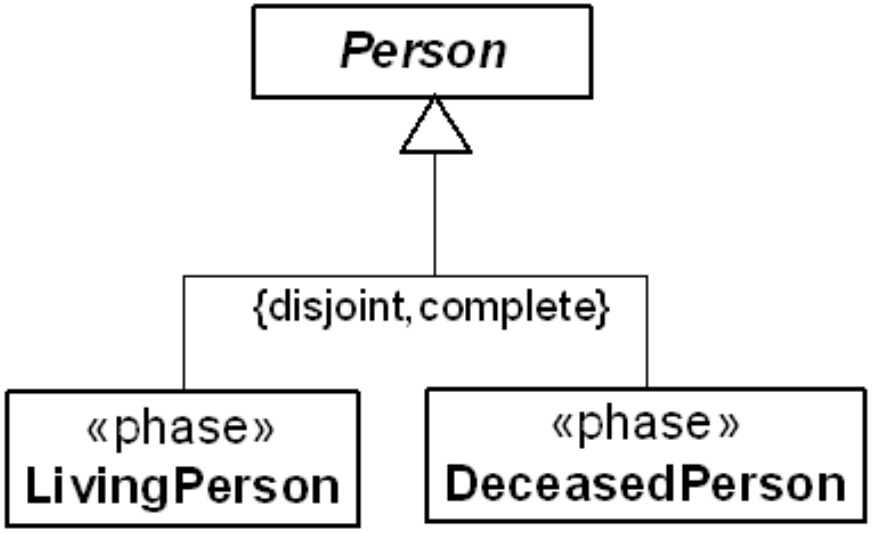
Student(x) =def Person(x) ∧ ∃y School(y) ∧ enrolled-at(x,y)



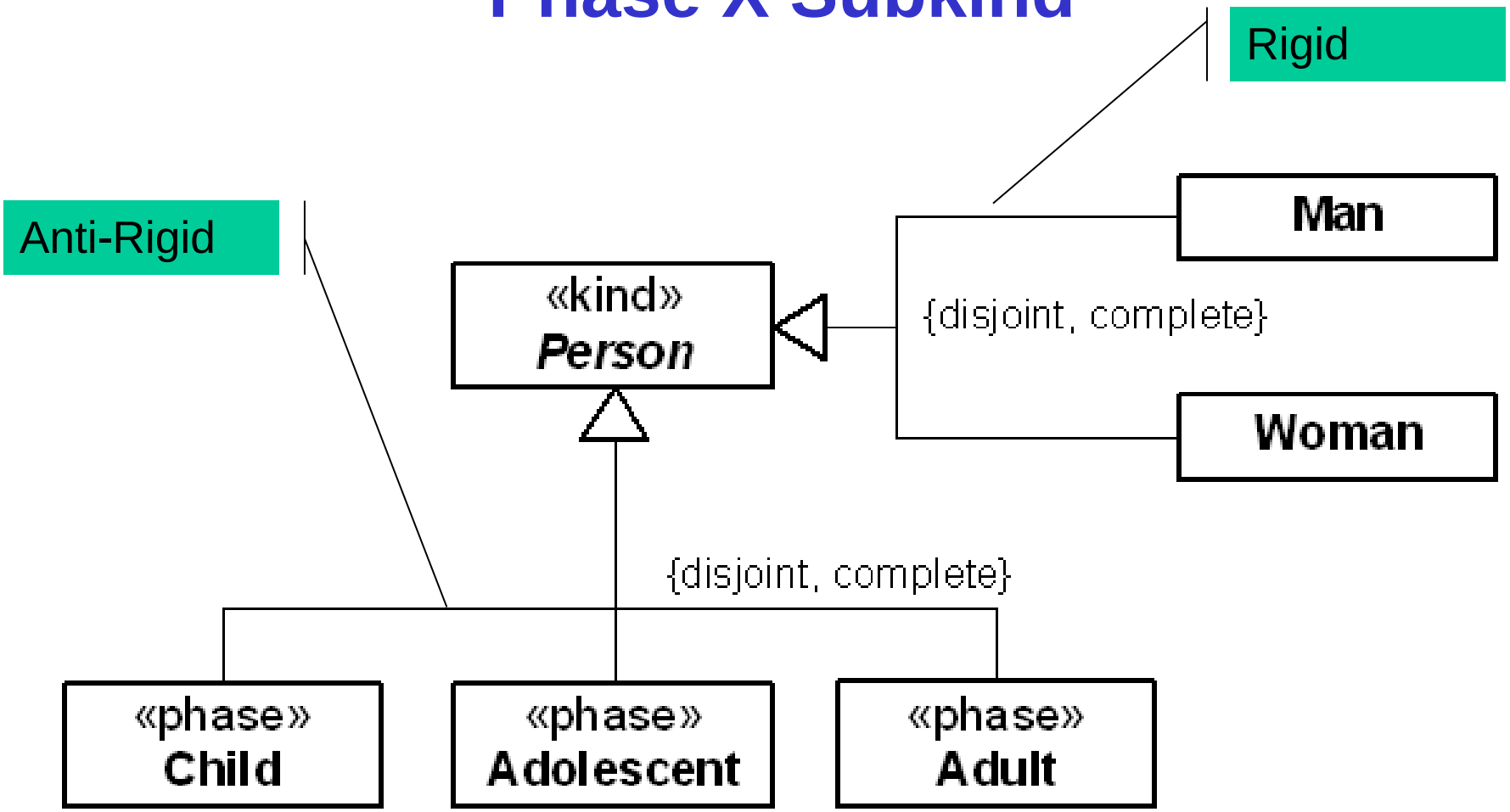
It must be a world in which *x* is not an instance of *Student*

Phases

- Na anti-rigid specialization of a kind based in an intrinsic property
- Phases define a phase partition, disjoint they and complete a generalization set



Phase X Subkind



Meta-Type Categories

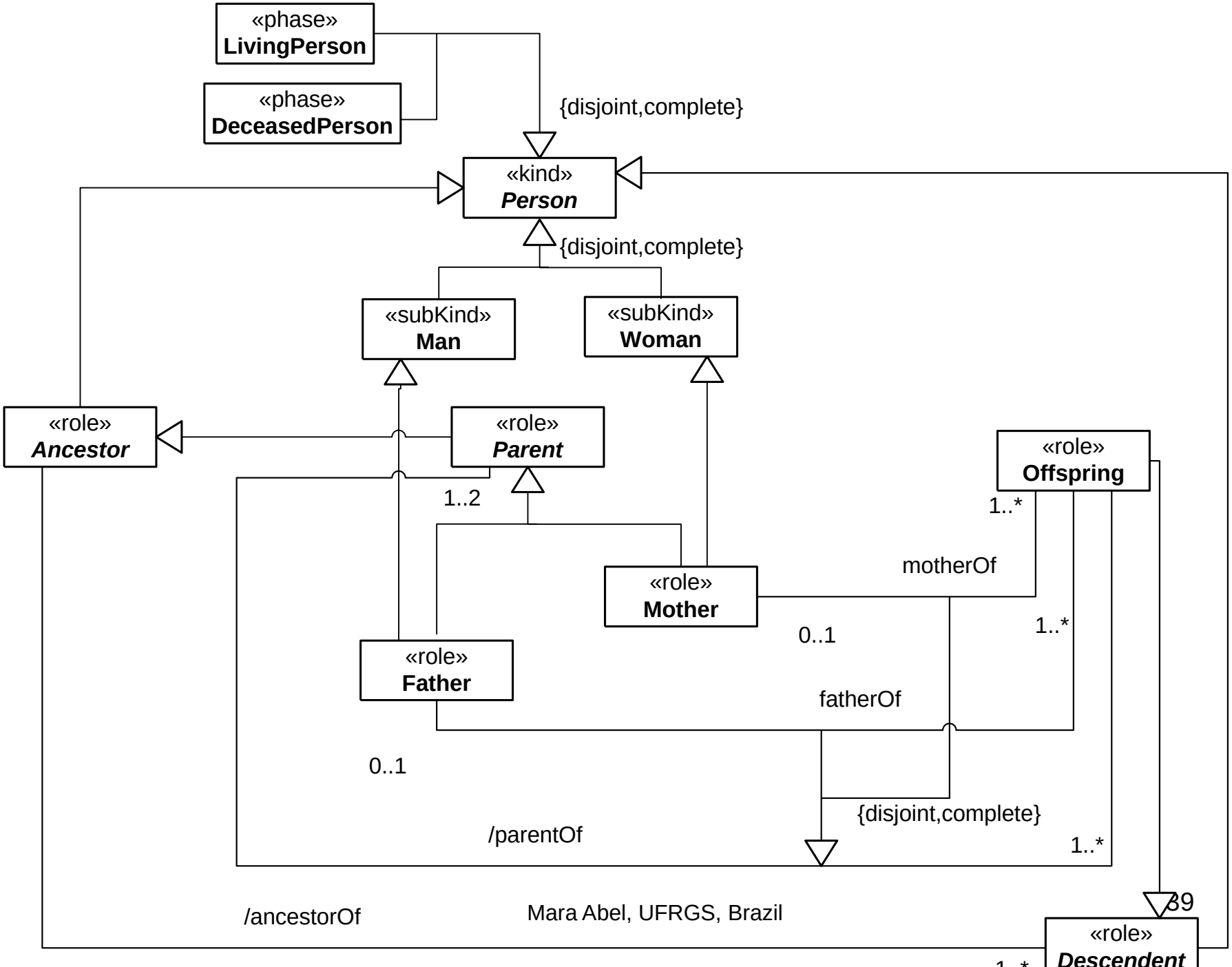
Meta-Type Categories	Provides Identity	Carries Identity	Rigidity	Relational Dependency
SORTAL	-	+	+/-	+/-
« kind »	+	+	+	-
« subkind »	-	+	+	-
« role »	-	+	-	+
« phase »	-	+	-	-
NON-SORTAL	-	-	+/-	+/-

Analysis

- Propose a conceptual model for genealogical relations.
- To model:
  - ***Person, woman, man, progenitor, mother, father, ancestor, descendant, offspring, alive, deceased.***

Check if each concept is:

- Kind,
- Subkind,
- Role,
- Phase

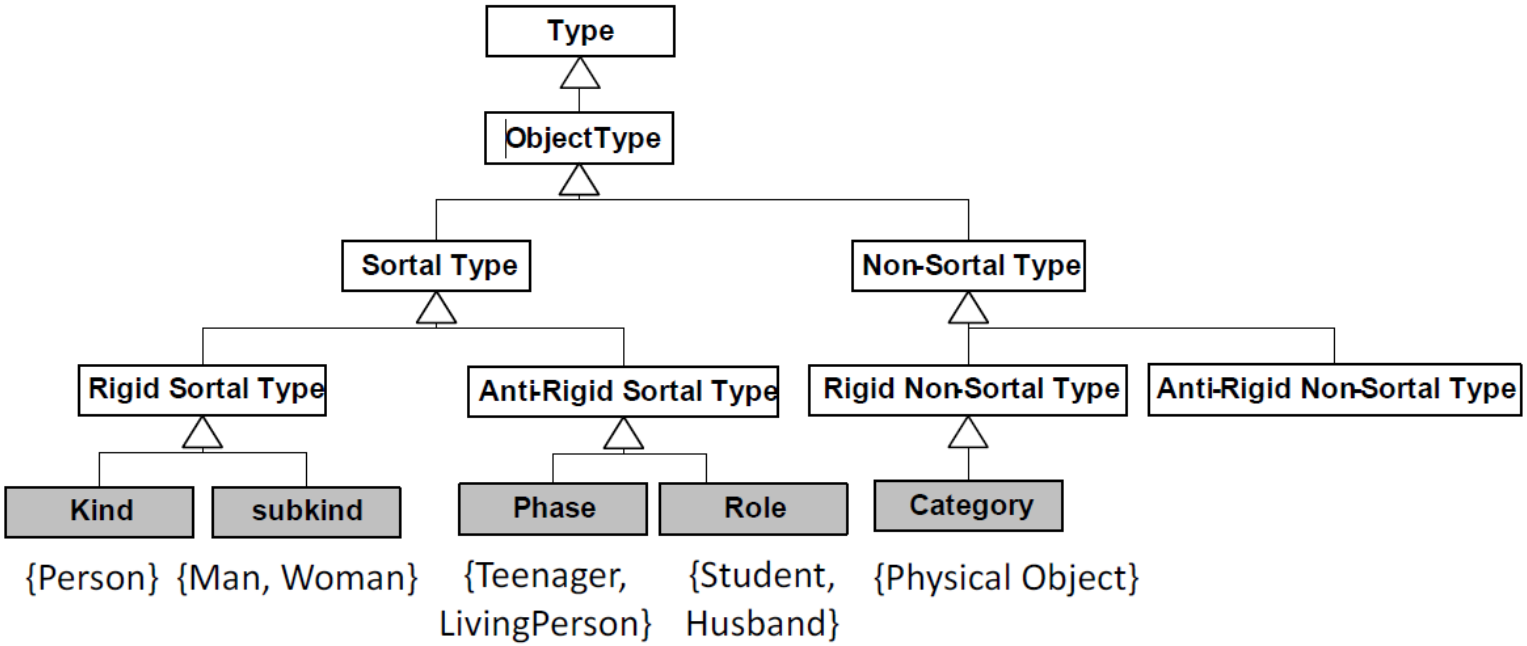

«phase»
LivingPerson
«phase»
DeceasedPerson
{disjoint,complete}
«kind»
Person
{disjoint,complete}
«subKind»
Man
«subKind»
Woman
«role»
Ancestor
«role»
Parent
«role»
Offspring
1..2
1..*
motherOf
«role»
Mother
0..1
1..*
«role»
Father
fatherOf
0..1
{disjoint,complete}
/parentOf
1..*
/ancestorOf
«role»
Descendent

Meta-Type Category

Meta-Type Categories	Provides Identity	Carries Identity	Rigidity	Relational Dependency
SORTAL	-	+	+/-	+/-
« kind »	+	+	+	-
« subkind »	-	+	+	-
« role »	-	+	-	+
« phase »	-	+	-	-
NON-SORTAL	-	-	+/~	+/-
« category »	-	-	+	-
« roleMixin »	-	-	-	+
« mixin »	-	-	~	-

Category (I- R+ DR-)

- Defines abstract classes, which means that the instances of categories are instances of other kind or subkind

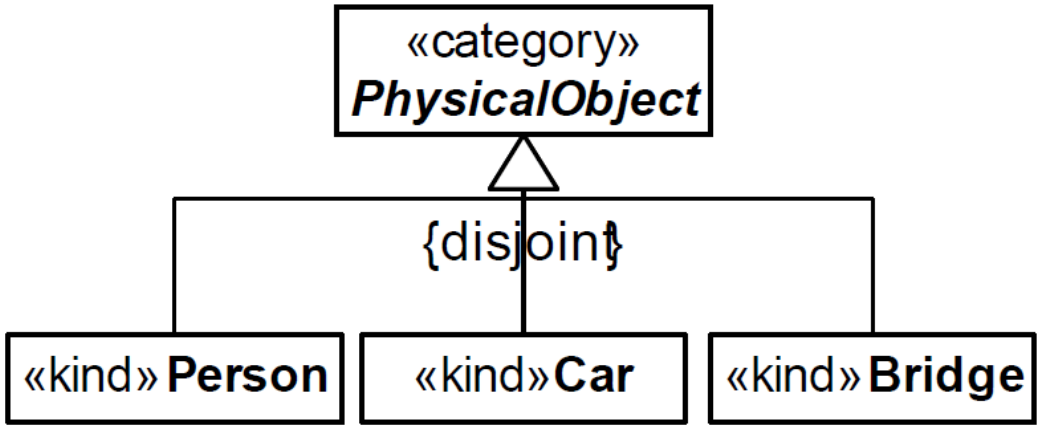


Category

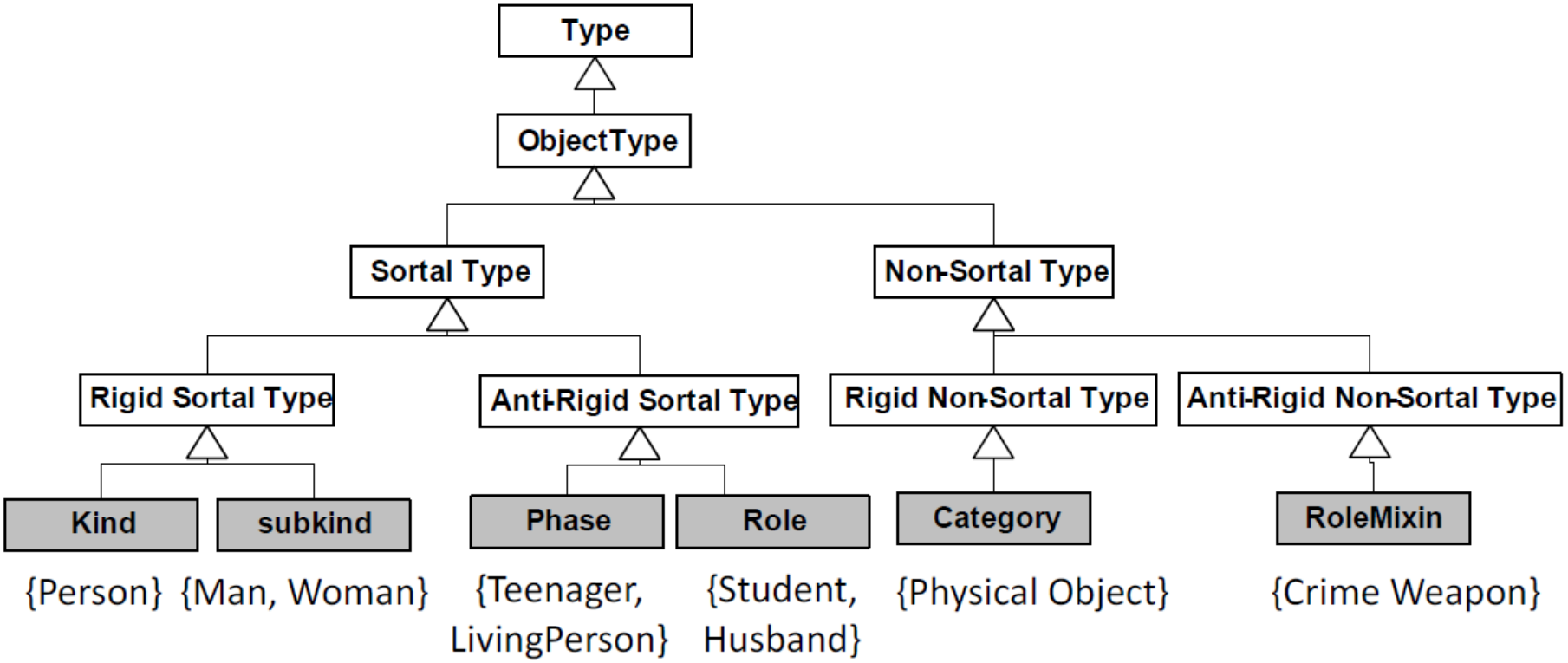
A kind can be subtype of multiple categories and a category can be a supertype of multiple kinds

Since all kinds are disjoint (otherwise the instances in their intersection would inherit multiple principles of identity), all subtypes of a category form a disjoint generalization set

On the other hand, since categories represent features of multiple kinds, they are very seldom complete



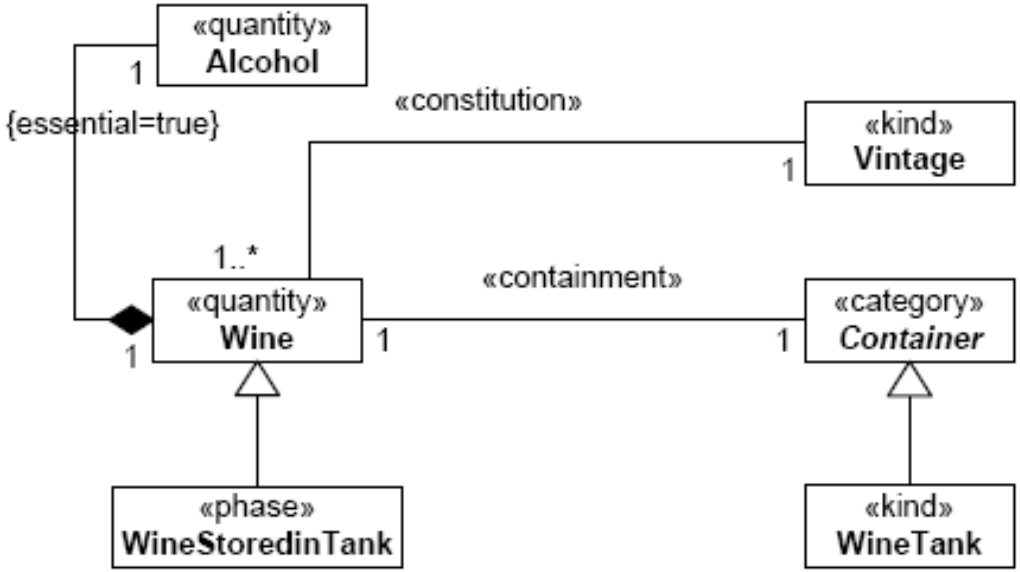
Role Mixin (I- R- D+)



Role Mixin (I- R- D+)

- Anti-rig and relational dependent non-sortals representing contingent features in modal sense of different kinds
- Defines abstract classes; the instances of role mixins are of distinct kinds
- RoleMixins cannot be super-type of a category or a kind.

Quantity



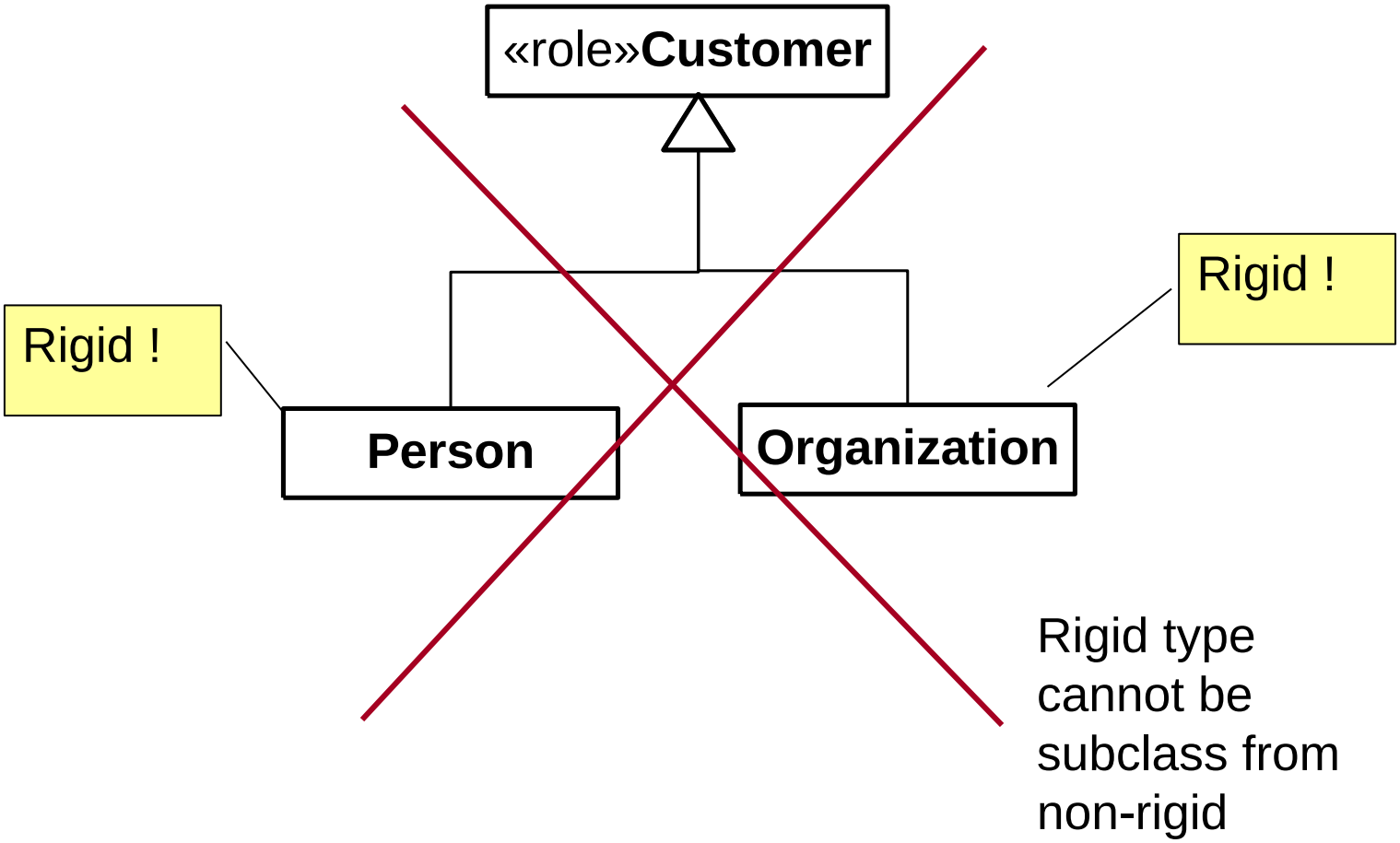
- Rigid, carries and supplies identity,
- Doesn't meet the individuation principle
- Uncountable

Analysis

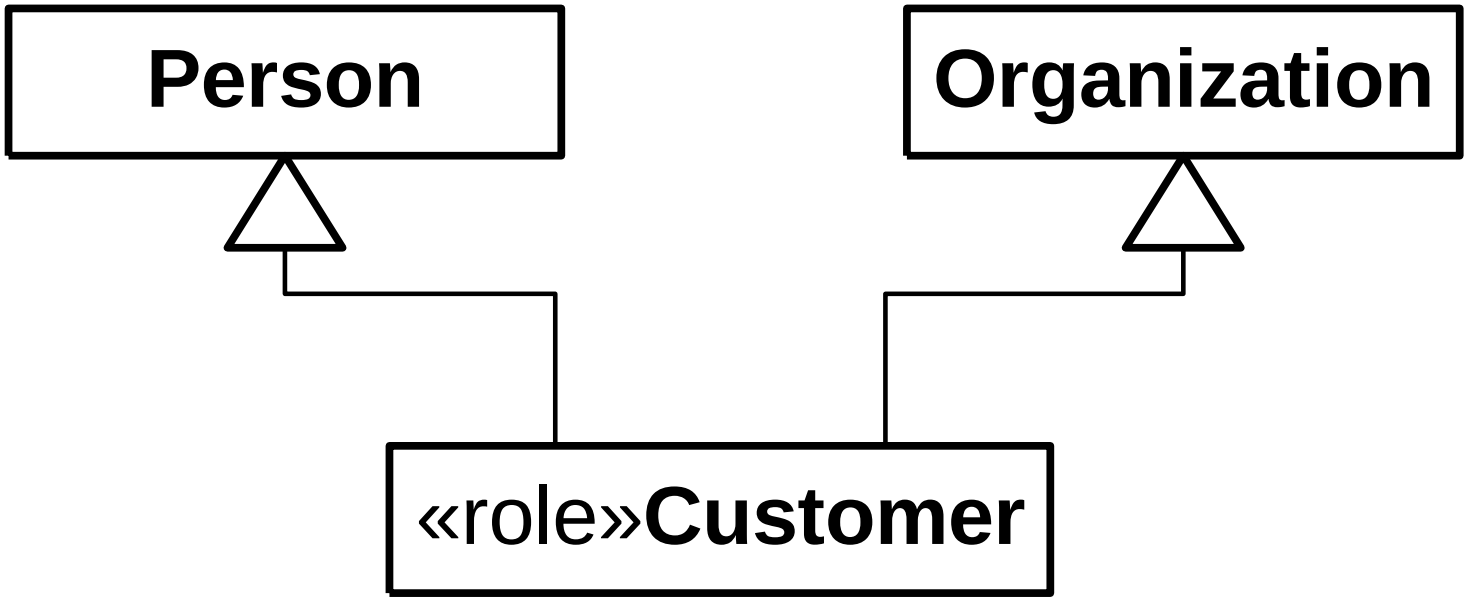
- For each concept analyze the metaproperties
  - R : Rigidity
  - U : Unity
  - I : Identity
  - O : Own identity
  - *D : Dependency*

Common Modeling Problems

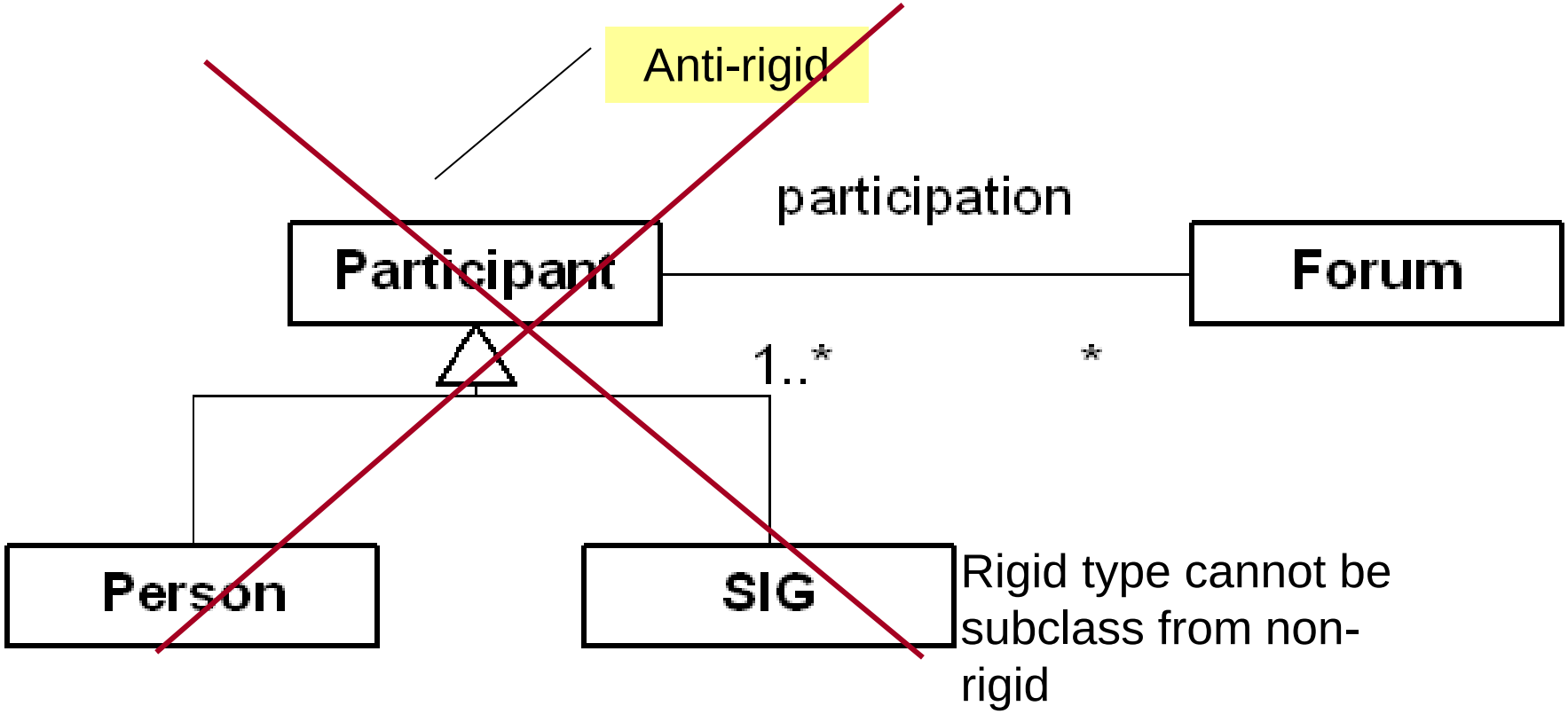
Roles with disjunct types



Roles with disjunct types

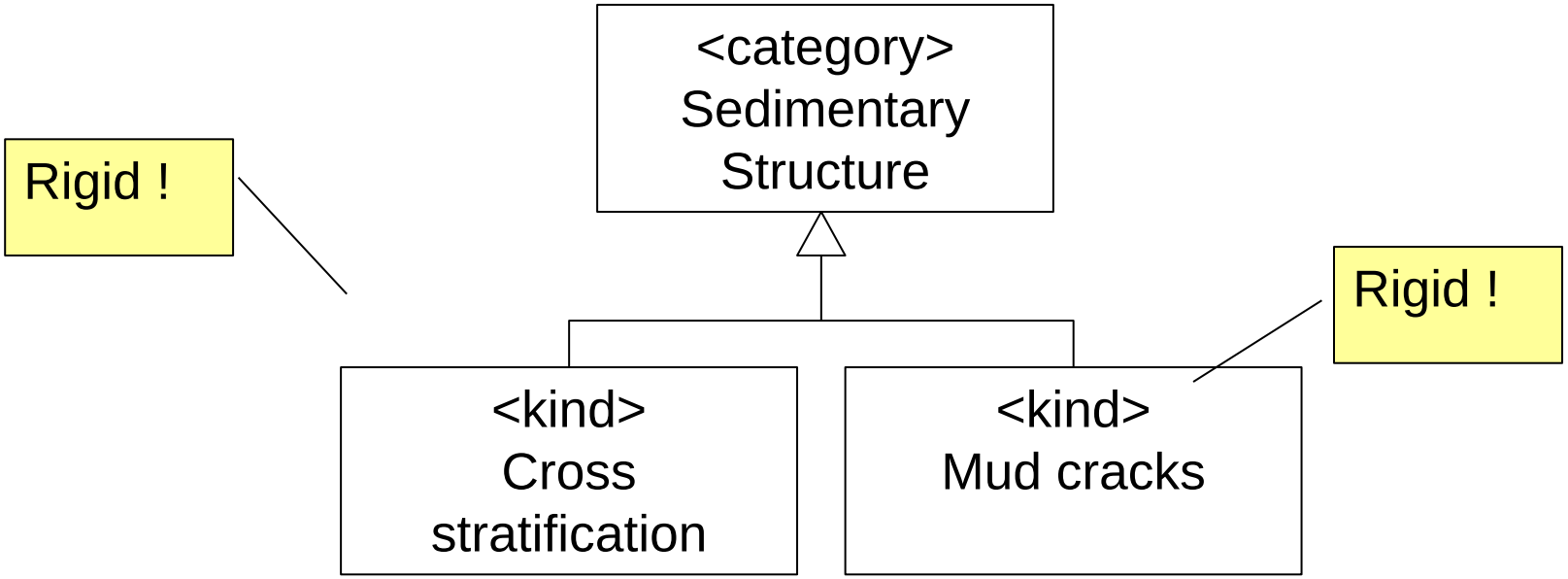


Another example



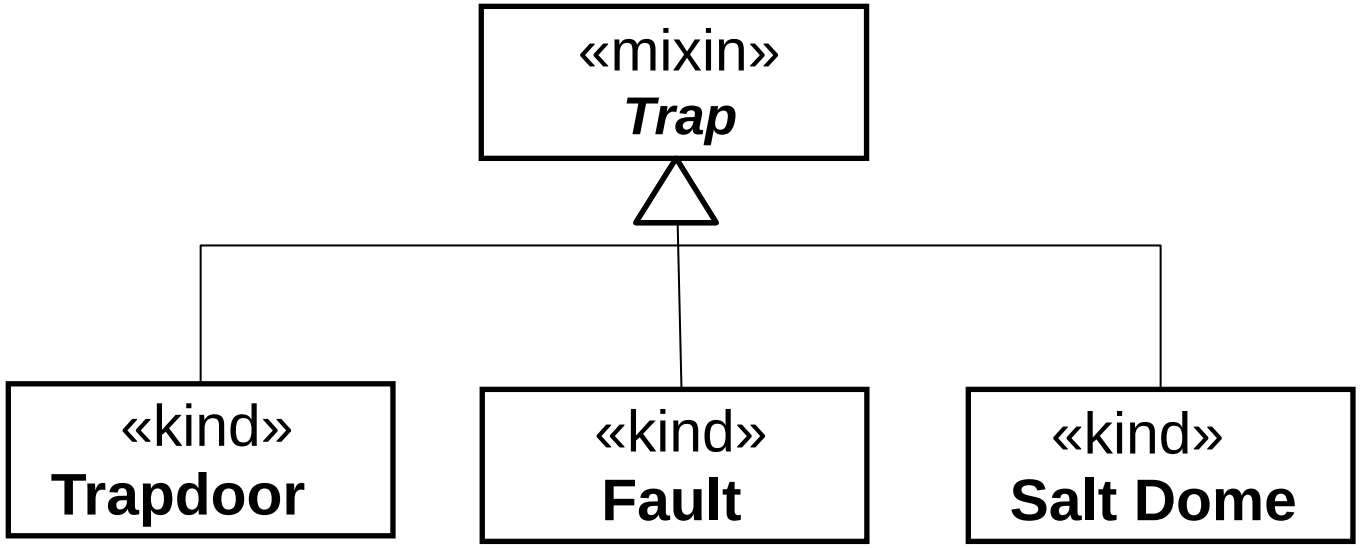
Rigid type cannot be subclass from non-rigid

Category



- A <Category> allows the creation of an abstraction of essential common properties from all instances of disjunct types. It is a rigid type.

Mixin



- A <Mixin> allows the creation of an abstraction of properties that are essential to some objects and not for others. It is a semi-rigid type.
- Porous media,

Summary of Universals

- Universals represent the abstraction of objects or representable entities of the world
- Individuais represent instances (existentials)
- Only rigid types (Kind, subkind, Collective, Quantity) have instances

Sortal Meta-Types

- **Kind** : Rigid, countable, carries and supplies identity, meets the individuation and countability principles
- **Sub-kind** : Rigid, countable, carries but does not supplies identity, meets the individuation and countability principles
- **Quantity** : Rigid, carries and supplies identity, does not meet the individuation nor countability principles, homeomorph
- **Collective** : Rigid, countable, may have identity or not
- **Phase** : Anti-rigid, countable, carries but doesn't supply identity
- **Role** : Anti-rigid, countable, carries but does not supply identity, relational dependent
- **Category** : Rigid, does not carry identity
- **RoleMixin** : Anti-rigid, does not carry identity, relational dependent
- **Mixin** : Anti-rigid, does not carry identity

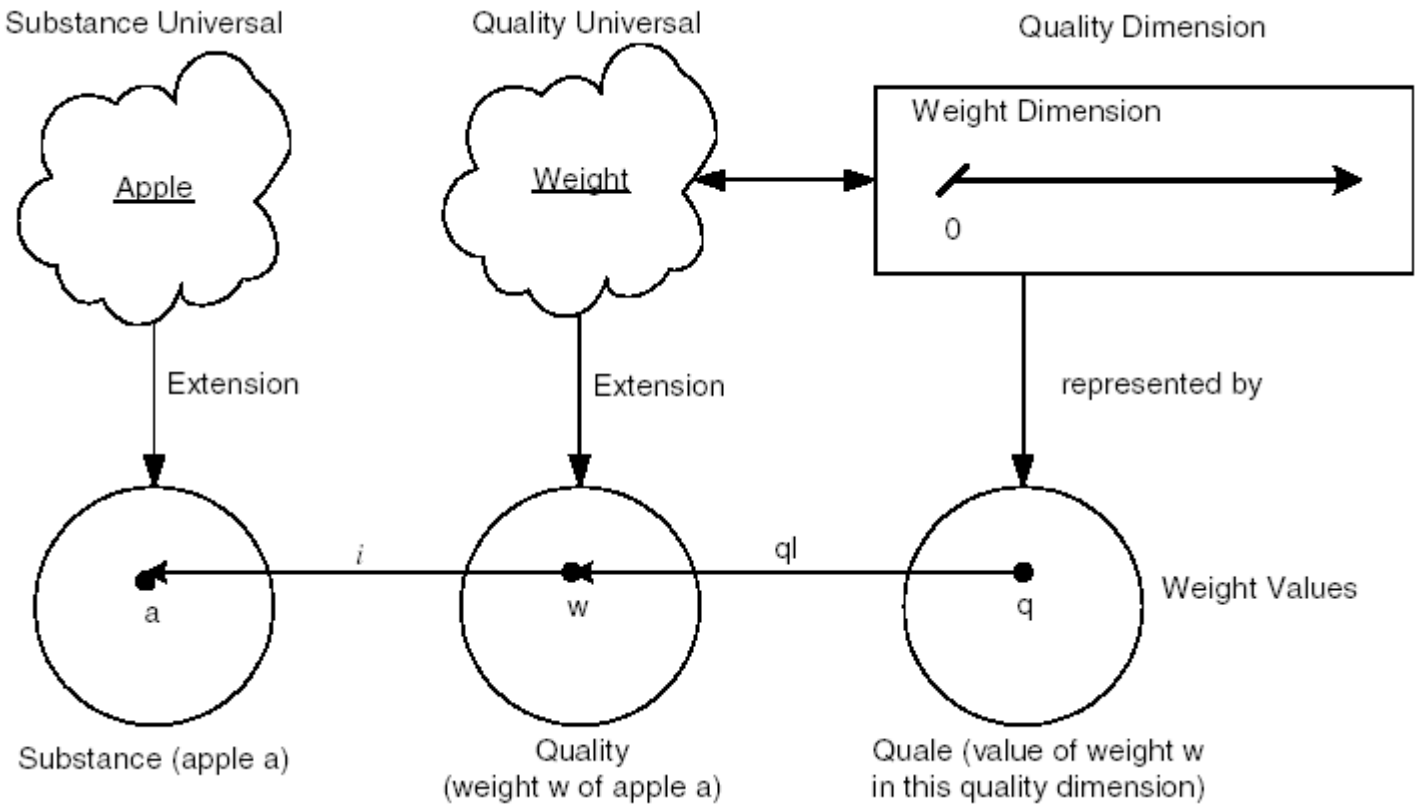
Meta Constructs

1. Sortals
2. Mixins
3. Momments
4. Relations

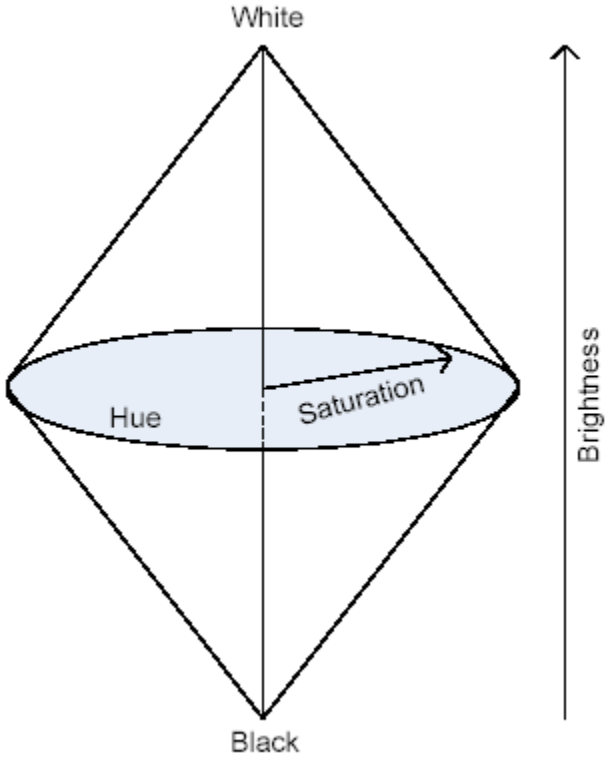
Qualia and Quality Dimension

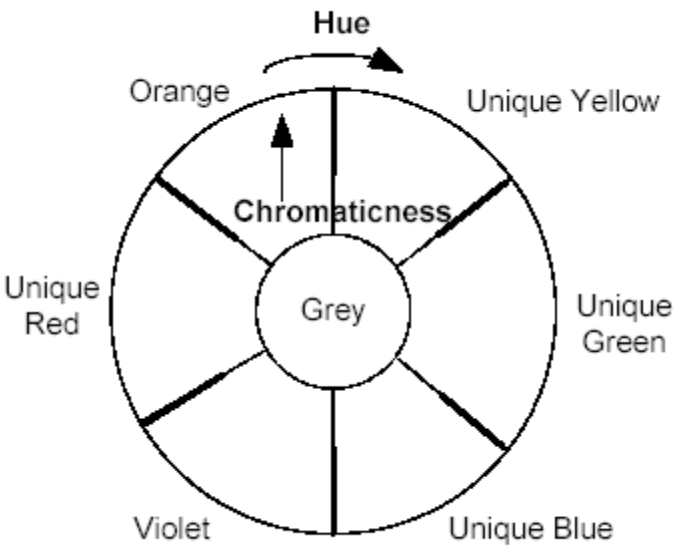
- The color of an apple (trope) is distinguished from its *value* (a particular red tone)
- The value of a trope is denominated *quale*, and describes that value's position in a quality dimension
- For each perceived or conceived property ther is an associated dimension in human cognition.
- For example, the WEIGHT property is associated with an unidimensional structure relative to the interval of natural numbers.

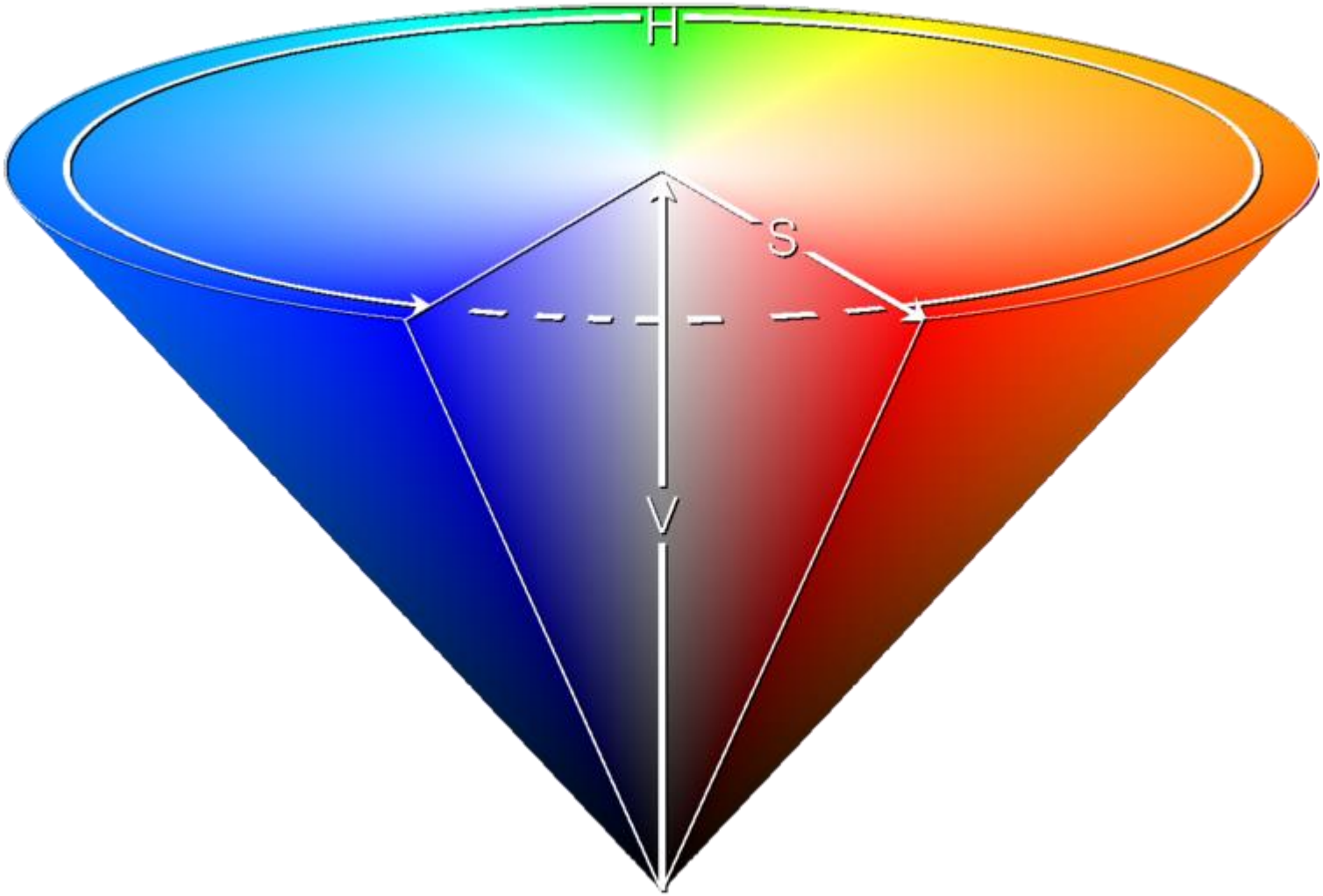
Qualia and Quality Dimension



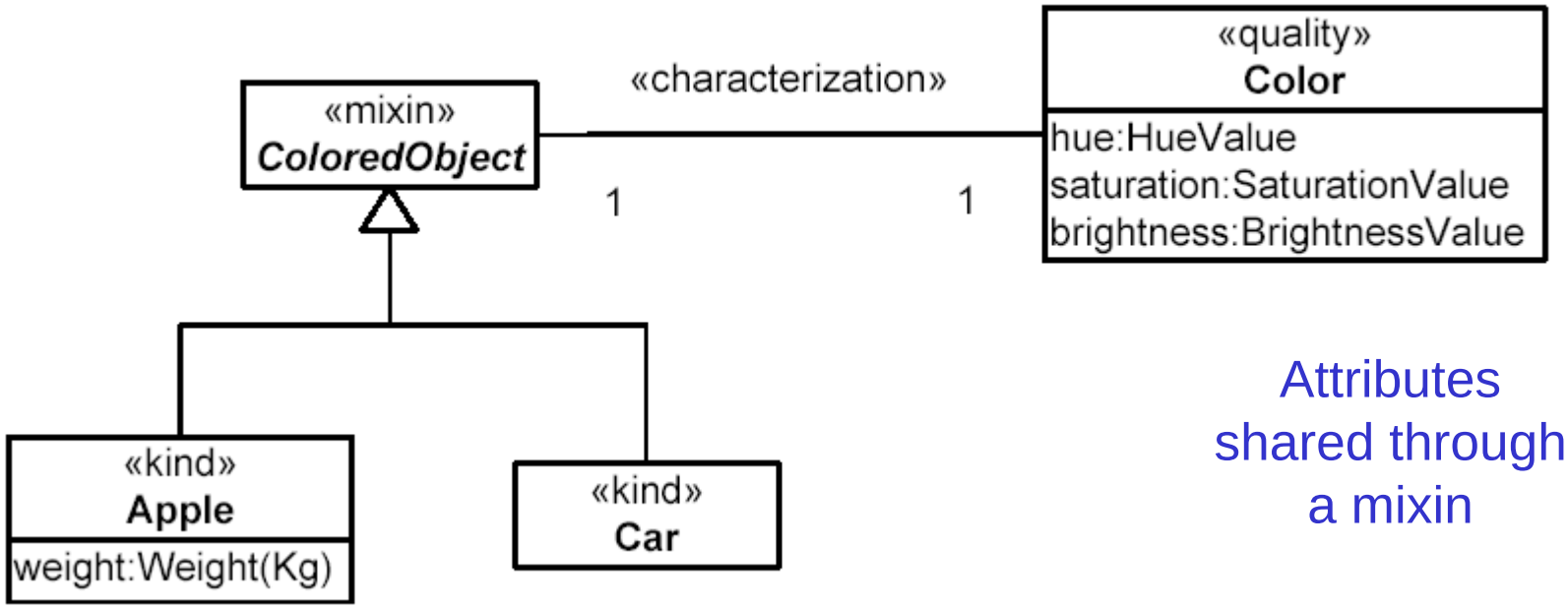
Quality Dimension Structure






H
S
V

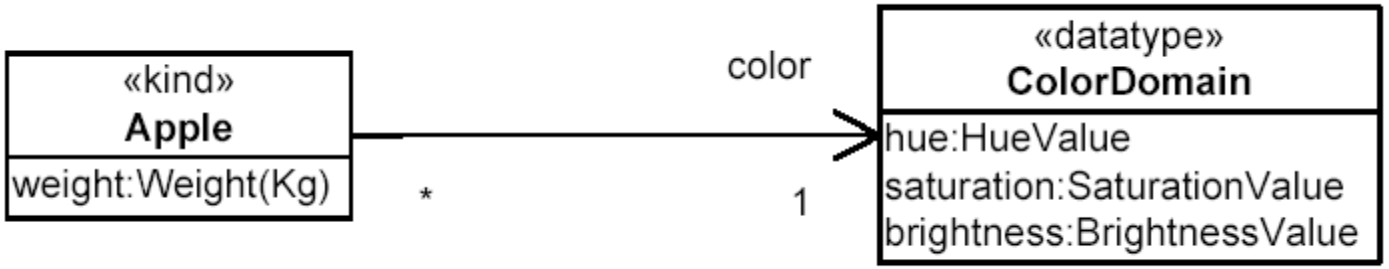
Quality Representation (1)



Attributes shared through a mixin

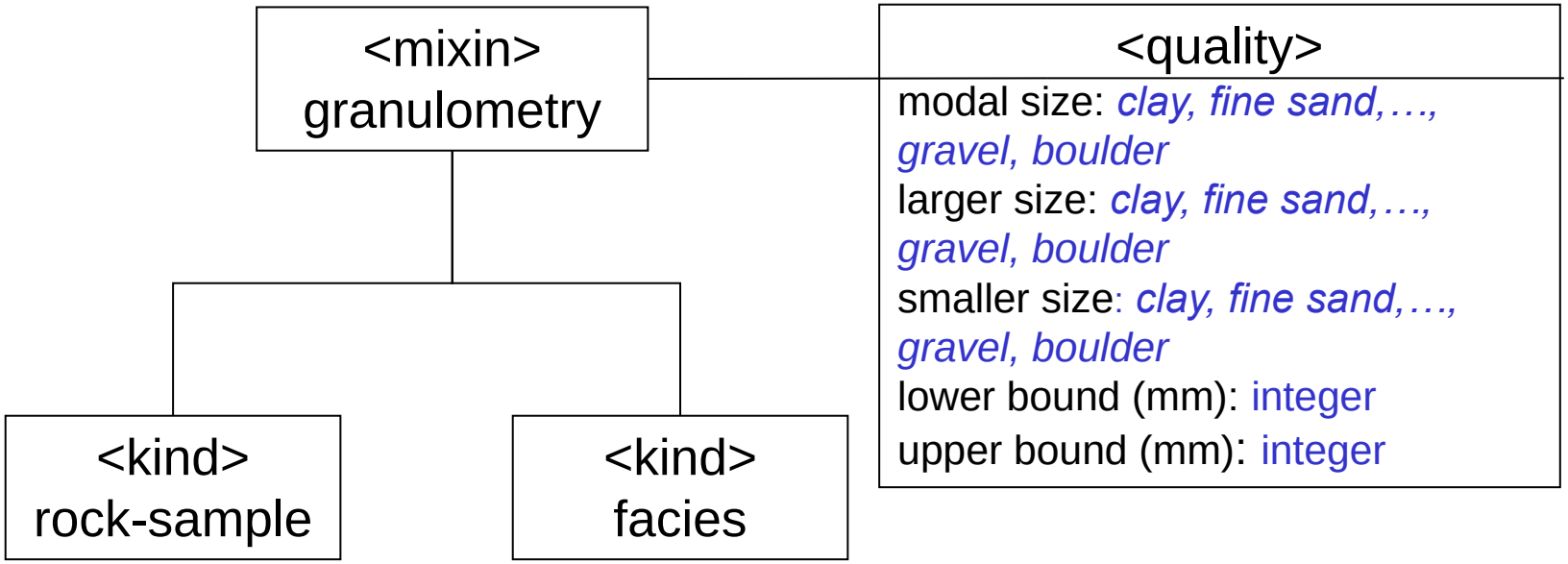
- The function of the attributes hue, saturation and brightness map each individual quality to the qualia in the quality domain that composes the color domain

Quality Representation (2)



- Hue, saturation and brightness attributes associated to the Universal itself

PETROLEDGE example



- List of sizes and their translations in a table shared by both knowledge bases

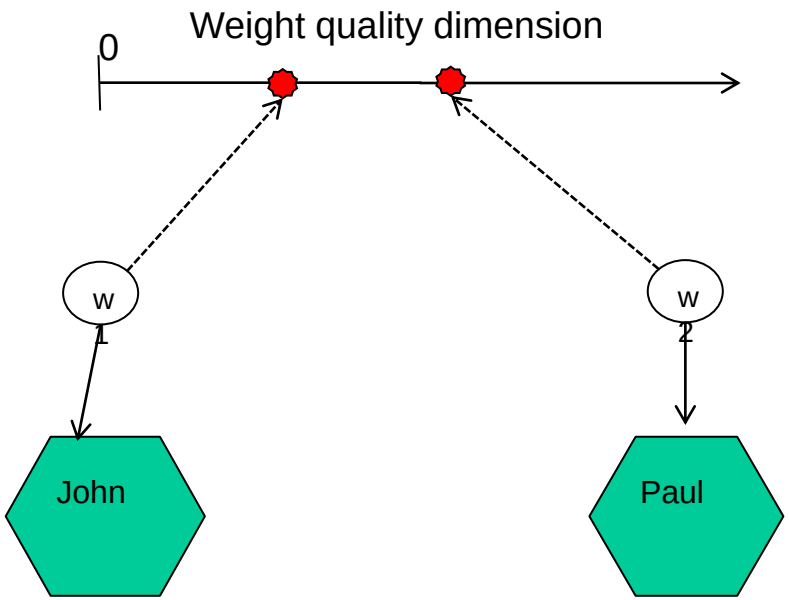
Relation

- An universal that depends existantially in two or more subtantial universals defines a relation

  kiss (john, mary)

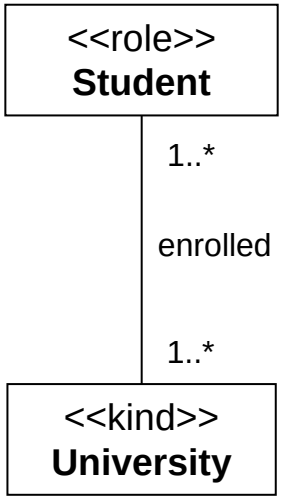
Types of Relations

Formal Relations

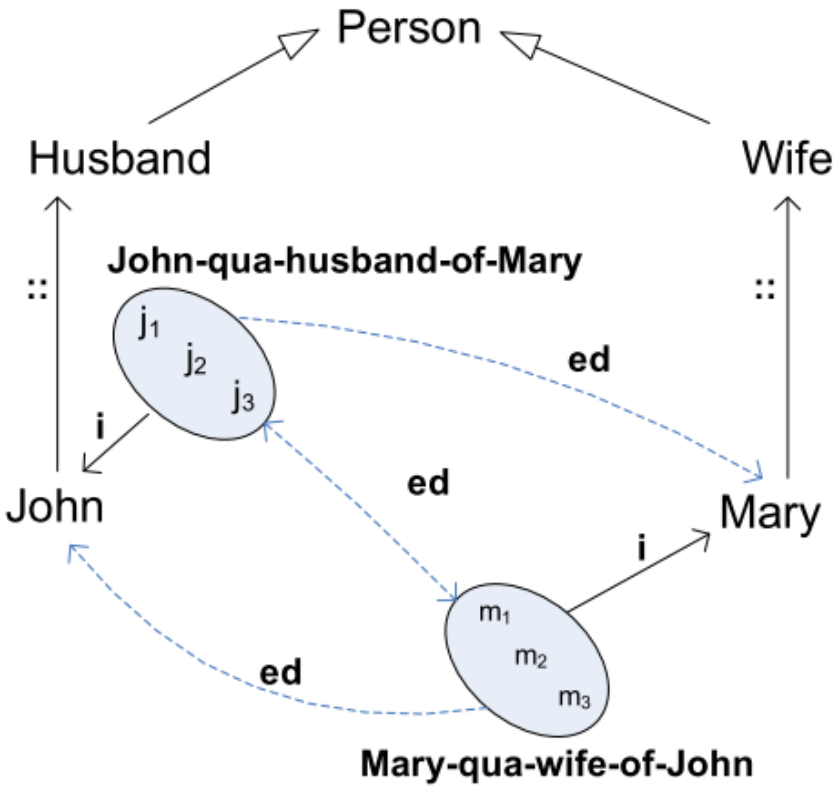


Heavier (Paul, John)?

Material Relations

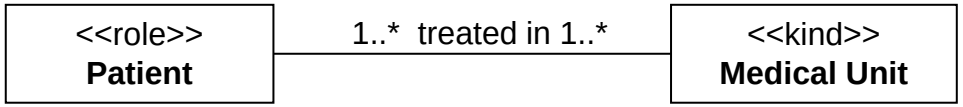


Relator Universals have Existential Dependency



There are *roles* of John and Mary, such as John as Mary's husband and Mary as John's wife.

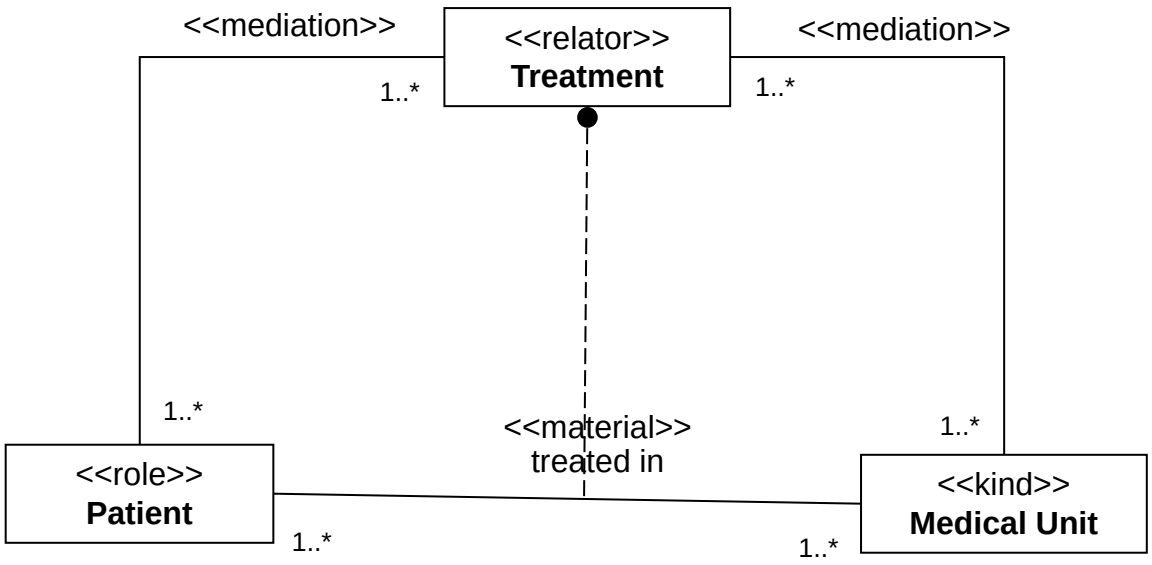
Material Relations



When the material relation is not materialized in an universal, there are problems in the interpretation of this relation cardinality

- In a treatment, one pacient is treated by multiple medical unity and receive many treatments
- In a treatment, one pacient is treated by multiple medical unity and receive many treatments
- In a treatment, many pacients are treated by one medical unity and receive many treatments
- In a treatment, one pacient is treated by one medical unity and receives many treatments
- ???
- Reflects on errors in queries

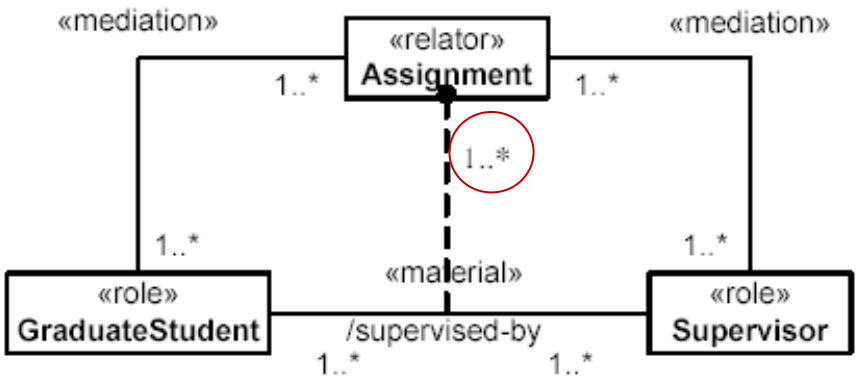
Material relations must be explicit represented

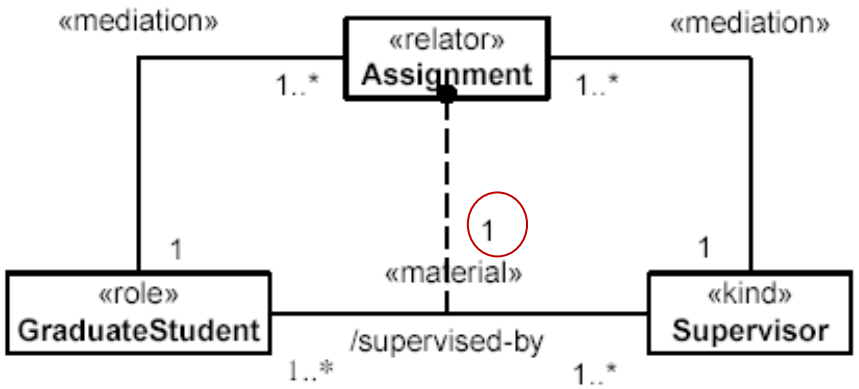


Use the <*relator*> universal

Material Relations – another example

- A student has onde or more supervisor and an supervisor may oversee one or more students

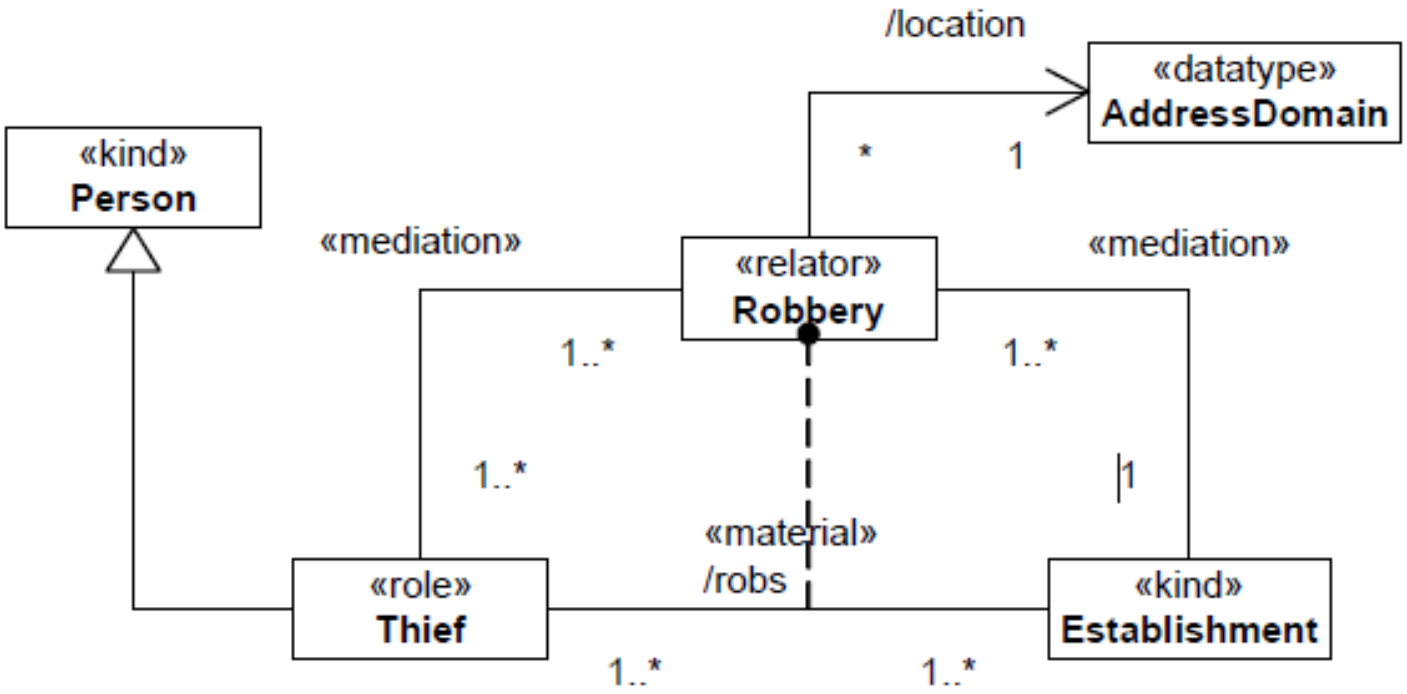




Only one assignment happen between student and supervisor

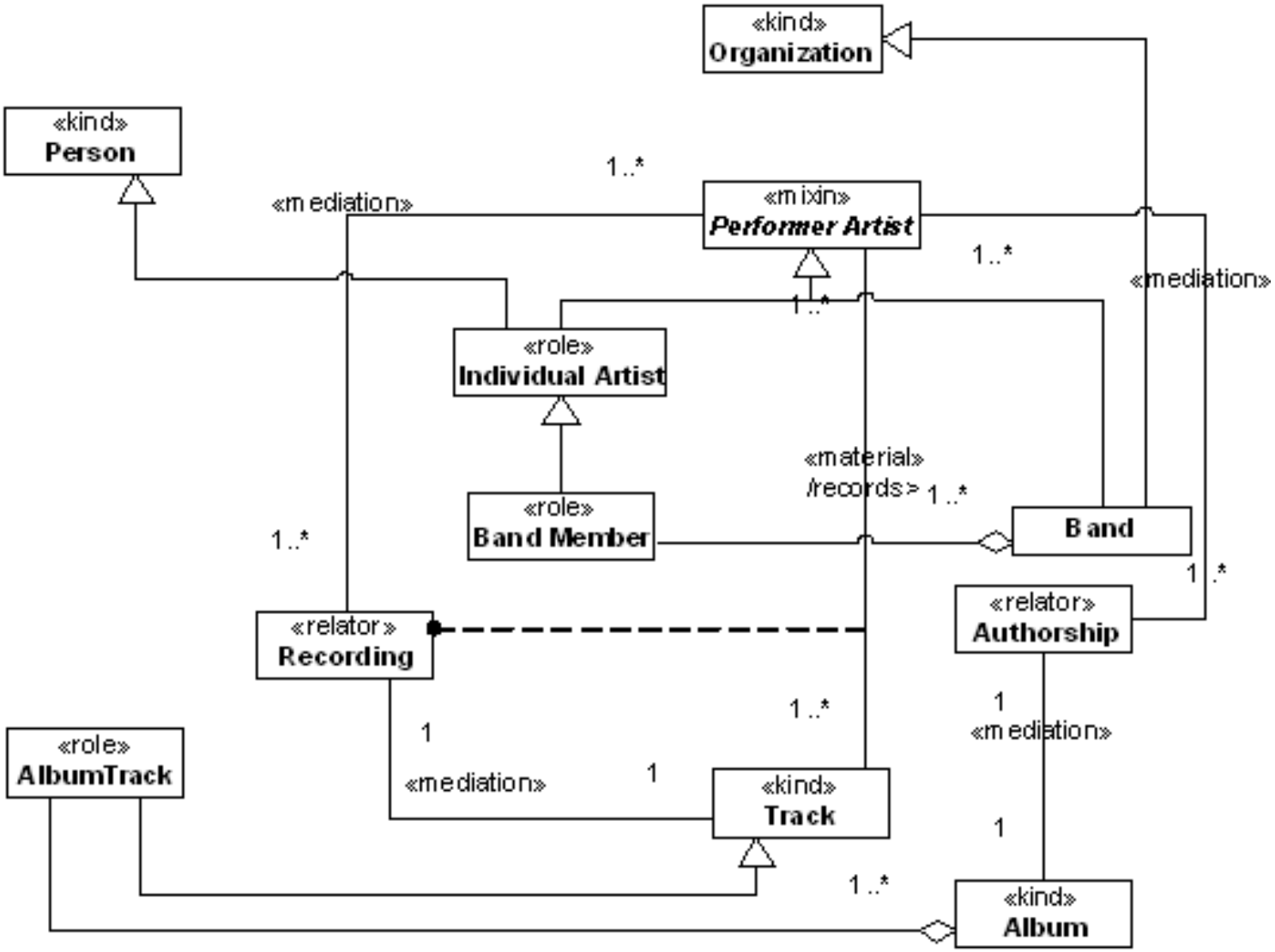
Properties of the Relations

- A relation may have associated qualities



Example

- Make a representation of the music industry domain
- Represent the artists, which may be persons or bands
- Bands are composed by persons
- Artists have albums with tracks recorded by artists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Performer Artist
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Track
1
1..*
«kind»
Album

Meta Constructs

1. Meta-types
2. Properties
3. Structuring Relations
   - 3.1 Part-Whole Relations
   - 3.2 Taxonomic Relations

Types of Partonomies

- Component/Object: hand/arm.
- Member/Collection: tree/forest.
- Portion/Mass: pedaço/bolo.
- Characteristic/Activity: pay/buy.
- Location/Area: Madrid/Spain.
- Constituent/Object: aluminum/plane.
- Phase/Process: teenager/growth.

Parte-Whole Relations

- SubQuantidatyOf
- SubCollectiveOf
- MemberOf
- SubComponentOf

Part-Whole Relations

- Primary features
  - Non-reflective, asymmetric, transitive (sometimes)
- Secondary features
  - Shareable, essential, inseparable, mandatory

Types of Partonomies

- SubQuantityOf
  - Holds between Quantities
  - Ex. alcohol is part of wine
  - **Always transitive**
- SubCollectiveOf (set partitioning)
  - Holds between Collectives
  - Ex. the northern part of the Black Forest is part of the Black Forest
  - **Always transitive**

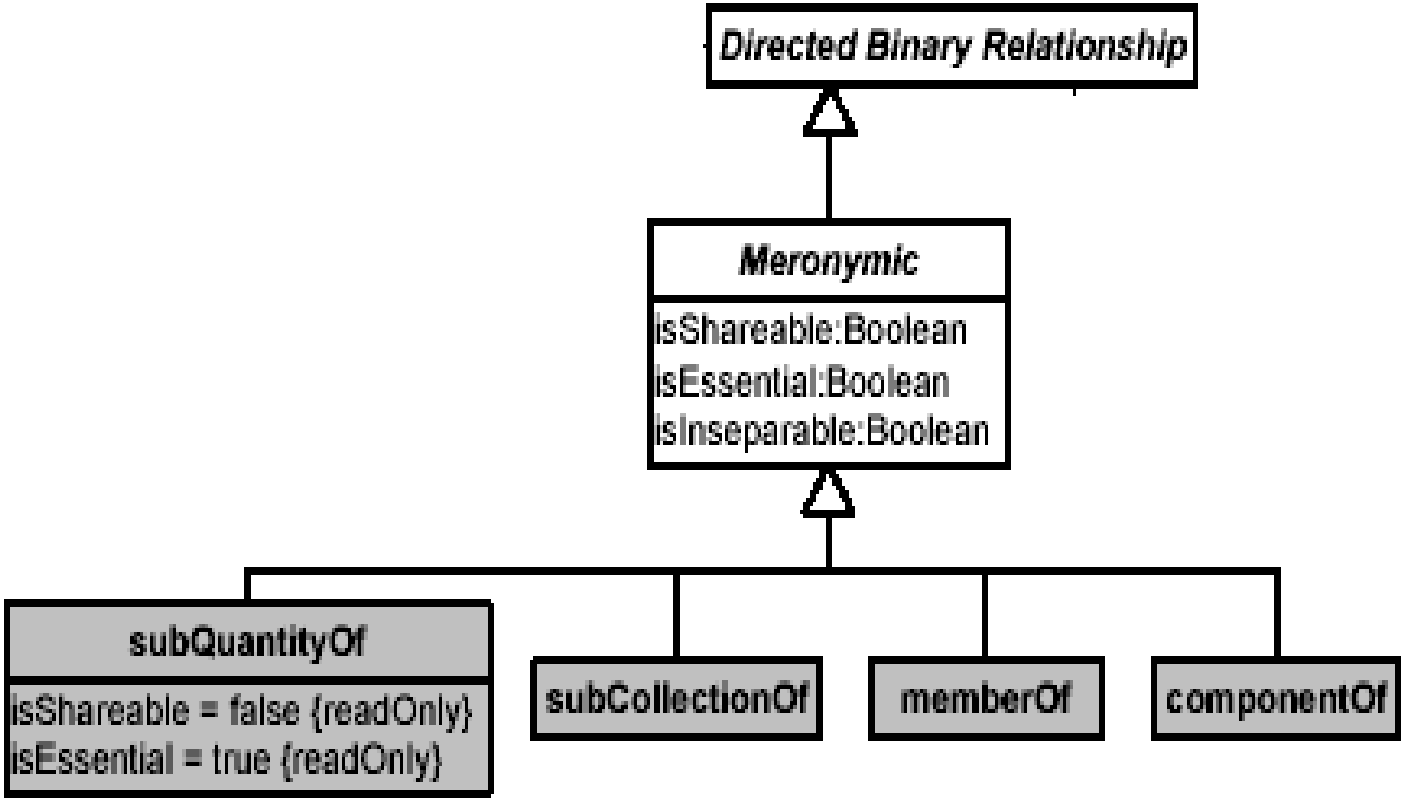
Types of Partonomies

- Member-Collective (*MemberOf*) (set element)
  - Holds between objects and their collectives
  - Ex. a card is part of the deck of cards
  - **Not transitive**
- The combination of *MemberOf* and *SubCollectiveOf* is **always transitive**
  - Ex. A tree is part of the northern part of the Black Forest, which is part of the Black Forest. This tree is part of the Black Forest.

Types of Partonomies

- Complex Functional Component (*FunctionalComponentOf*)
  - Holds between complex wholes
  - E.g., an engine is a functional component of a car, a heart is a functional component of the circulatory system
  - **Not transitive** (transitive in some cases and not in others)

Types of Part-Whole Relations



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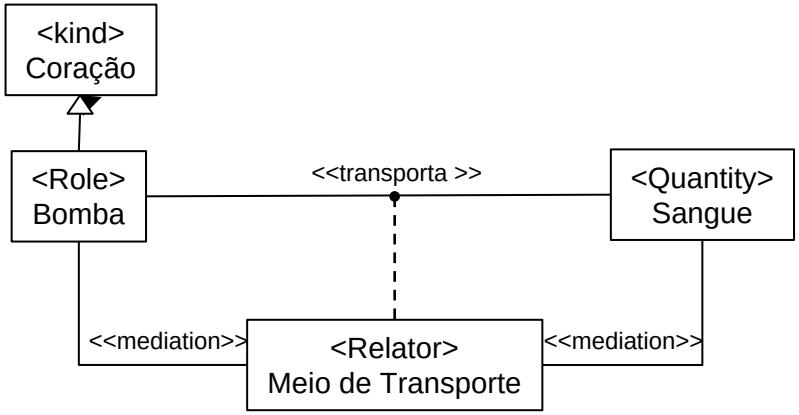
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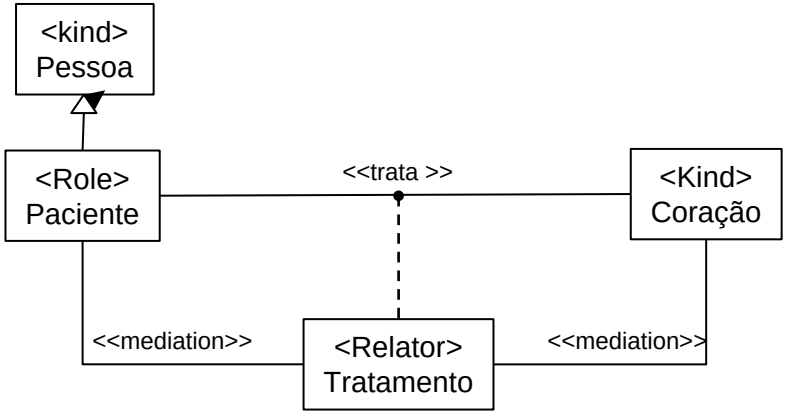
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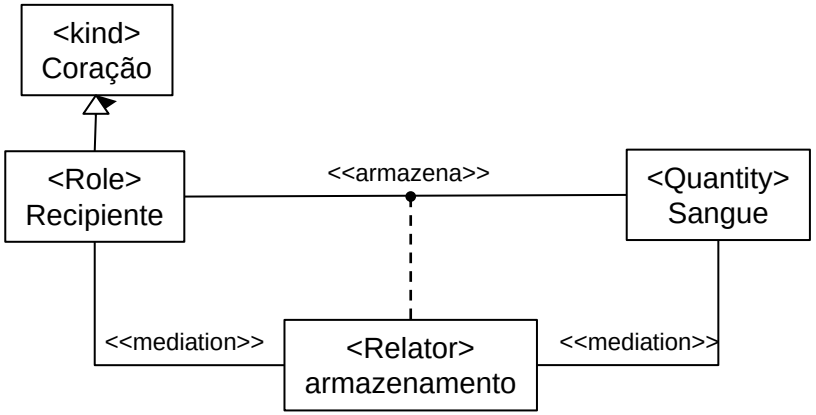
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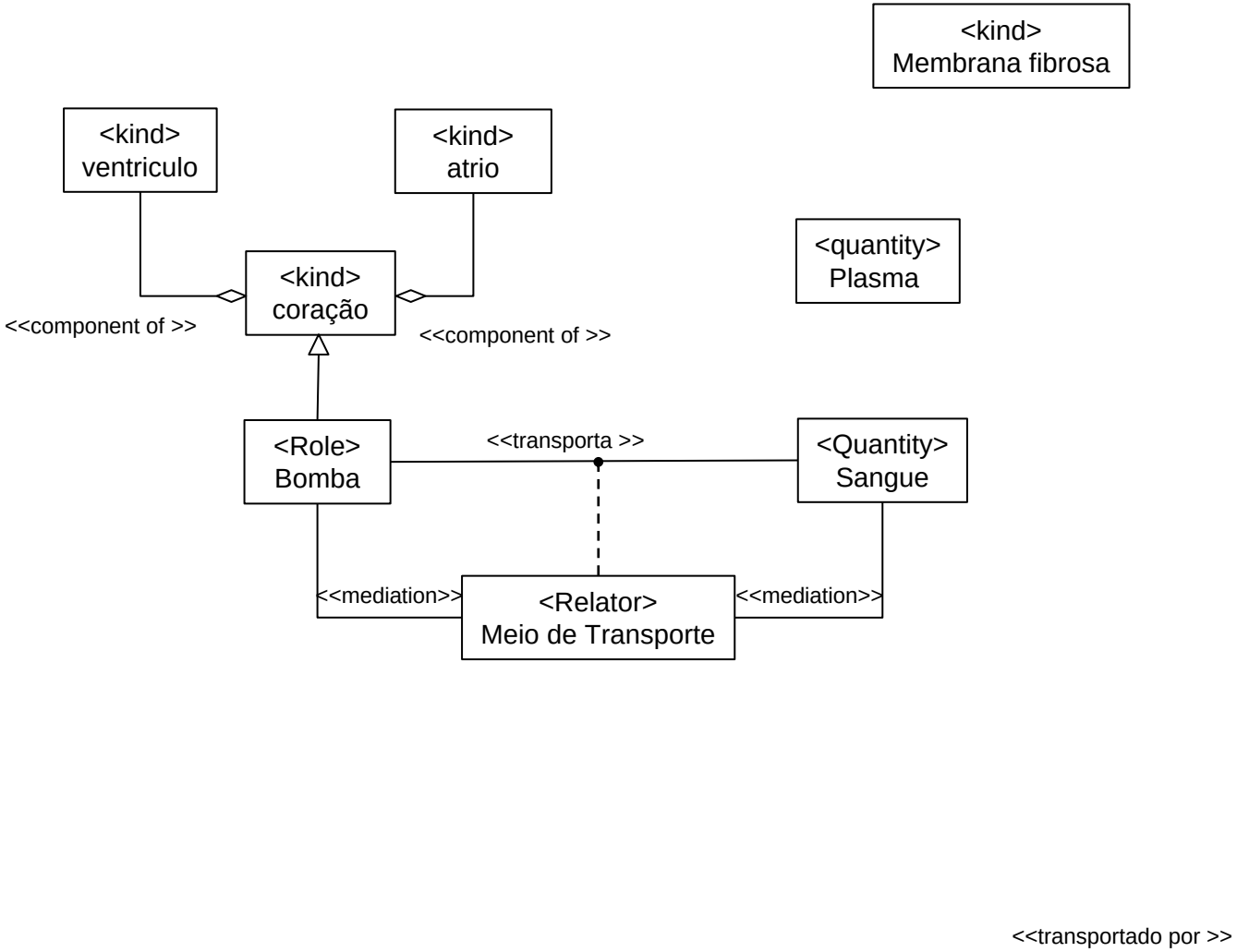
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